# 45. Octal Serial Peripheral Interface (OSPI)

This is the OSPI_B version of the OSPI peripheral module.

Flash memory ISSI IS25WX064-JWLE is packaged for SIP product.

OSPI_B is referred to as OSPI in this chapter.

## 45.1 Overview

The xSPI (eXpanded Serial Peripheral Interface) protocol specifies the interface for nonvolatile memory devices, which provides high data throughput, low signal count, and limited backward compatibility with legacy SPI devices. The electrical interface can deliver up to 333 MB per second raw data throughput. The OSPI is compliant with JEDEC standard JESD251 (profiles 1.0 and 2.0), JESD251-1 and JESD252.

JESD251 specifies two interface profiles where profile 1.0 is Octal SPI and profile 2.0 is HyperBus™ (HyperRAM™ and HyperFlash™). OSPI supports the QSPI protocol.

Table 45.1 lists the OSPI specifications, Figure 45.1 shows a block diagram, and Table 45.2 lists the I/O pins.

**Table 45.1 OSPI specifications**

| Item | Description |
|---|---|
| Number of units | 2 units |
| Protocol | Compliant with the xSPI protocol*1 |
| Data transmission and reception | Issue transaction for up to 2 slaves as masters.<br>Only one of the memory devices can operate at a time. |
| Transfer speed | Support the transfer at xSPI333 |
| Mode | ● Support the following protocol modes:<br>– 1-, 4-, 8-pin with SDR/DDR (1S-1S-1S, 4S-4D-4D, 8D-8D-8D)<br>– 2-, 4-pin with SDR (1S-2S-2S, 2S-2S-2S, 1S-4S-4S, 4S-4S-4S)<br>● Configurable address length<br>● Configurable initial access latency cycle<br>● Support XiP mode |
| OSPI function | ● Support write data mask<br>● Support in-band reset<br>● Memory-mapping<br>– Unit 0 supports up to 256 MB address space each CS<br>– Unit 1 supports up to 128 MB address space each CS<br>– 2 control channels per unit for the bus masters<br>• ch 1: GLCDC1 bus master<br>• ch 0: Other bus master<br>– Prefetch function for burst-read with low latency<br>– Outstanding buffer for burst-write with high throughput<br>● Manual command<br>– Configurable up to 4 commands<br>– Status register polling function<br>● Input strobe port timing shift |
| Decryption function | Decryption on-the-fly is available for memory-map read |
| Interrupt source | ● Error interrupt<br>● Completion interrupt |
| Module-stop function | Module-stop state can be set to reduce power consumption |
| TrustZone Filter | ● Security attribution can be set for the IO register area<br>● External address space is defined as non-secure |

Note 1. The OSPI is compliant with JEDEC standard JESD251 (profiles 1.0 and 2.0), JESD251-1 and JESD252.

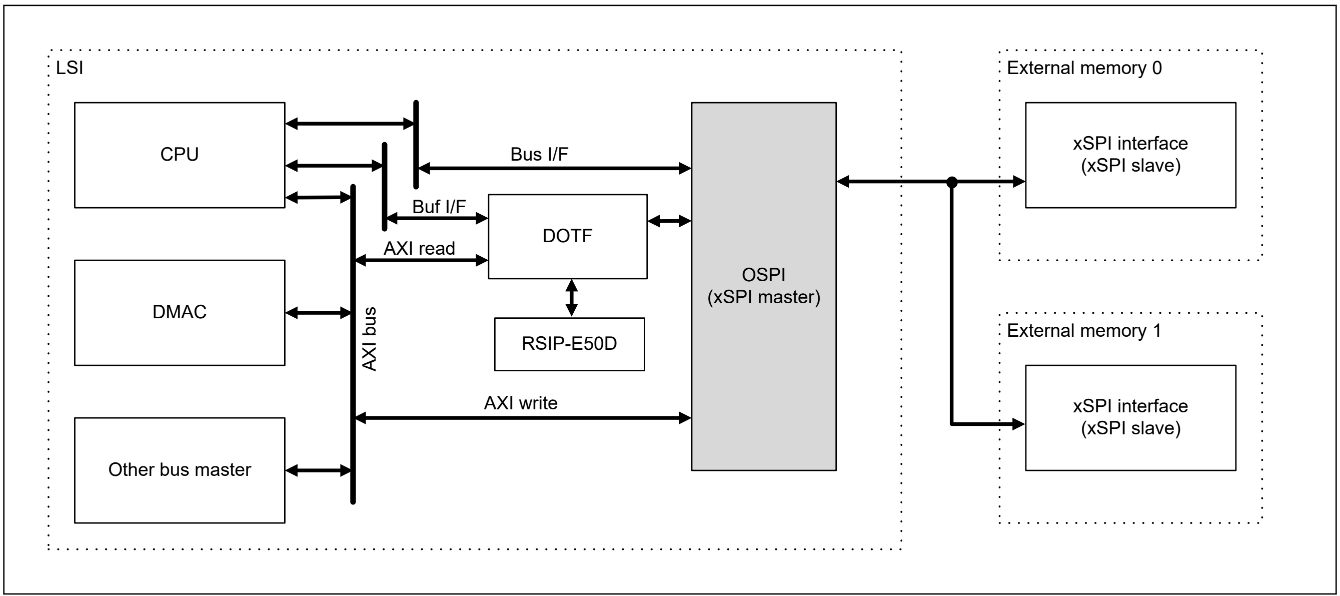


**Figure 45.1 Block diagram**

**Table 45.2 OSPI I/O pins**

| Pin name | I/O | Function |
|---|---|---|
| OM_n_SCLK | Output | Clock positive |
| OM_n_SCLKN | Output | Clock negative |
| OM_n_CS0 | Output | Chip selection for slave 0 |
| OM_n_CS1 | Output | Chip selection for slave 1 |
| OM_n_DQS | I/O | Read data strobe/write data mask |
| OM_n_SIO0 | I/O | Data 0 input/output |
| OM_n_SIO1 | I/O | Data 1 input/output |
| OM_n_SIO2 | I/O | Data 2 input/output |
| OM_n_SIO3 | I/O | Data 3 input/output |
| OM_n_SIO4 | I/O | Data 4 input/output |
| OM_n_SIO5 | I/O | Data 5 input/output |
| OM_n_SIO6 | I/O | Data 6 input/output |
| OM_n_SIO7 | I/O | Data 7 input/output |
| OM_n_RESET | Output | Master reset status for slave 0, 1 |
| OM_n_RSTO1 | Input | Slave reset status for slave 1 |
| OM_n_ECSINT1 | Input | Interrupt for slave 1/error correction status for slave 1 |
| OM_n_WP1 | Output | Write protect for slave 1 |

Note: For OM_n_SIO7-0, OM_n_SCLK, OM_n_SCLKN and OM_n_DQS pins, 30Ω ±1% resistor must be put on the board to comply with JESD251 I/O driver definition. Renesas recommends this for the operation with proper signal quality.

Note: n = 0, 1

## 45.2 Register Descriptions

### 45.2.1 OSPI Configuration Registers

These registers configure the xSPI Master function. These registers should be configured in the initialization phase before issuing xSPI transactions.

### 45.2.1.1 WRAPCFG : OSPI Wrapper Configuration Register

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x000

| Bit position: | 31 | | | 28 | | | | 24 | | | | 20 | | | | 16 | | | | 12 | | | | 8 | | | | 4 | | | | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | DSSFTCS1[4:0] | | | | | — | — | — | — | — | — | — | — | — | — | — | DSSFTCS0[4:0] | | | | | — | — | — | — | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 7:0 | — | These bits are read as 0. The write value should be 0. | R/W |
| 12:8 | DSSFTCS0[4:0] | DS Shift for Slave 0<br>These bits configure the number of delay cells for an OM_DQS port. A delay cell is used to adjust the DS sampling timing. When automatic calibration is enabled, it can be updated automatically. In this case, it should not be written by the user.<br>0x00: No shift<br>0x01: Add a delay of one cell<br>⋮<br>0x1E: Add a delay of 30 cells<br>0x1F: Add a delay of 31 cells | R/W |
| 23:13 | — | These bits are read as 0. The write value should be 0. | R/W |
| 28:24 | DSSFTCS1[4:0] | DS Shift for Slave 1<br>The function is the same as one of slave 0. | R/W |
| 31:29 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

The WRAPCFG register has functions to configure the xSPI Master function.

### 45.2.1.2 COMCFG : OSPI Common Configuration Register

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x004

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | OENEGEX | OEASTEX |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | ARBMD[1:0] | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 1:0 | ARBMD[1:0] | Channel Arbitration Mode<br>These bits select the behavior when system bus accesses from ch0 and ch1 to slave devices occurred simultaneously. It is used only for memory-mapping mode.<br>0 0: Round-robin (ch0-ch1-ch0-ch1…)*1<br>0 1: Always ch0 win<br>1 0: Always ch1 win*1<br>1 1: Reserved | R/W |
| 15:2 | — | These bits are read as 0. The write value should be 0. | R/W |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 16 | OEASTEX | Output Enable Asserting Extension<br>This bit extends one cycle output enable of data and DS during output enable asserting.<br>When set to 1, CS asserting should be extended (LIOCFGCSn.CSASTEX = 1 (n = 0, 1)).<br>This bit is used for case with latency cycle because OSPI output data can be conflicted with OSPI input data.<br>0: Do not extend one cycle output enable<br>1: Extend one cycle output enable | R/W |
| 17 | OENEGEX | Output Enable Negating Extension<br>This bit extends one cycle output enable of data and DS during output enable negating.<br>This bit is used for case with latency cycle because OSPI output data can be conflicted with OSPI input data.<br>0: Do not extend one cycle output enable<br>1: Extend one cycle output enable | R/W |
| 31:18 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3
Note 1. ch1 can only be used for GLCDC1 bus master.

The COMCFG register has functions to configure the xSPI Master function.

### 45.2.1.3 BMCFGCHn : OSPI Bridge Map Configuration Register Chn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x008 + 0x004 × n

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | CMBTIM[7:0] | | | | | | | | — | — | — | — | — | — | — | PREEN |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | MWRSIZE[7:0] | | | | | | | | MWRCOMB | — | — | — | — | — | — | WRMD |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | WRMD | System Bus Write Response Mode<br>This bit selects the timing of a system bus write response in memory-mapping mode. When set to 1, this bit returns the response after transmitting a frame on the xSPI bus.<br>When the system bus write response mode is enabled, the memory write combination mode must be disabled.<br>0: Return response after storing to the internal write buffer<br>1: Return response after issuing a write transaction to the xSPI bus | R/W |
| 6:1 | — | These bits are read as 0. The write value should be 0. | R/W |
| 7 | MWRCOMB | Memory Write Combination Mode<br>This bit selects to combine the OSPI data in write access of memory-mapping mode. When set to 0, the OSPI data size depends on the burst type and size of the system bus. When this field is set to 1, the data size depends on the MWRSIZE[7:0] field.*1<br>When set to 1, any write transaction can be temporarily held in this xSPI master.<br>0: Disable combination mode<br>1: Enable combination mode | R/W |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 15:8 | MWRSIZE[7:0] | Memory Write Size<br>These bits select the size to combine incremental address in memory-mapping mode. It transmits an xSPI frame with the data combined up to the configured size while the address is incremental. When a nonincremental address or a read transaction is detected before reaching the target size, it transmits the pending data into the xSPI bus.<br>0x00: Combine incremental address of up to 4 bytes<br>0x01: Combine incremental address of up to 8 bytes<br>⋮<br>0x0E: Combine incremental address of up to 60 bytes<br>0x0F: Combine incremental address of up to 64 bytes<br>0xFF: Combine incremental address of up to 2 bytes<br>Others: Setting prohibited | R/W |
| 16 | PREEN | Prefetch Enable<br>This bit enables the prefetch function for a read transaction in memory-mapping mode. It can reduce the latency for read transaction with incremental address.<br>0: Disable prefetch function<br>1: Enable prefetch function | R/W |
| 23:17 | — | These bits are read as 0. The write value should be 0. | R/W |
| 31:24 | CMBTIM[7:0] | Combination Timer<br>These bits specify the expiration period of a combination timer. This timer is counted by PCLKA.<br>0x00 means disabling the combination timer. When the timer is expired, the data in the combination buffer is pushed to the memory device. | R/W |

Note: S-TYPE-3, P-TYPE-3

Note 1. Writing to OSPI memory space other than 64 bit is prohibited during combination mode.

The BMCFGCHn register has functions to configure the xSPI Master function.

### 45.2.1.4 CMCFG0CSn : OSPI Command Map Configuration Register 0 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x010 + 0x010 × n

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | ADDRPCD[7:0] | | | | | | | | ADDRPEN[7:0] | | | | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | ARYAMD | WPBSTMD | ADDSIZE[1:0] | | FFMT[1:0] | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 1:0 | FFMT[1:0] | Frame Format<br>These bits configure xSPI frame format in memory-mapping mode.<br>See Table 45.7 for detail.<br>0 0: Normal format:<br>Command 1 byte, address ADDSIZE, data up to system bus transaction<br>0 1: 8D-8D-8D profile 1.0 format:<br>Command 2 bytes, address ADDSIZE, data up to system bus transaction<br>1 0: 8D-8D-8D profile 2.0 Command Modifier format:<br>Command & Modifier 6 bytes, data up to system bus transaction<br>1 1: 8D-8D-8D profile 2.0 Commands with Extended Command Modifier format:<br>Command & Modifier 6 bytes, data up to system bus transaction | R/W |

<table>
<tr><th>Bit</th><th>Symbol</th><th>Function</th><th>R/W</th></tr>
<tr><td>3:2</td><td>ADDSIZE[1:0]</td><td>Address Size<br>These bits configure the number of address byte in memory-mapping mode.<br>For 8D-8D-8D profile 2.0, configure to 4 bytes.<br>0 0: 1 byte (256-byte address space)<br>0 1: 2 bytes (64-KB address space)<br>1 0: 3 bytes (16-MB address space)<br>1 1: 4 bytes (4-GB address space)</td><td>R/W</td></tr>
<tr><td>4</td><td>WPBSTMD</td><td>Wrapping Burst Mode<br>When this bit is set to 1, the wrapping boundary between system bus access and xSPI memory should be matched.<br>0: Separate xSPI transfer at the wrapping address boundary<br>1: Do not separate xSPI transfer at the wrapping address boundary</td><td>R/W</td></tr>
<tr><td>5</td><td>ARYAMD</td><td>Array Address Mode<br>When this bit is set to 1, address for memory is mapped as {A[25:10], A[9:4], 6{RSV}, A [3:0]} where A[25:0] is normal address, and RSV is reserved value(0b). This bit is valid only when FFMT = 1d.<br>0: Normal address mode<br>1: Array address mode</td><td>R/W</td></tr>
<tr><td>15:6</td><td>—</td><td>These bits are read as 0. The write value should be 0.</td><td>R/W</td></tr>
<tr><td>23:16</td><td>ADDRPEN[7:0]</td><td>Address Replace Enable<br>These bits select the bits to replace the MSByte of system bus address in memory-mapping mode.<br>0: No replacement (xSPI frame address field is the same as system bus address)<br>1: Replacement
<table>
<tr><td>ADDSIZE[1:0]</td><td>Replace target bits of system bus address</td></tr>
<tr><td>00</td><td>bit 7 to bit 0</td></tr>
<tr><td>01</td><td>bit 15 to bit 8</td></tr>
<tr><td>10</td><td>bit 23 to bit 16</td></tr>
<tr><td>11</td><td>bit 31 to bit 24</td></tr>
</table></td><td>R/W</td></tr>
<tr><td>31:24</td><td>ADDRPCD[7:0]</td><td>Address Replace Code<br>These bits configure the code to replace the MSByte of system bus address in memory-mapping mode. It replaces the corresponding bits when the Address Replace Enable bit is set to 1.</td><td>R/W</td></tr>
</table>

Note: S-TYPE-3, P-TYPE-3

The CMCFG0CSn register has functions to configure the xSPI Master function.

### 45.2.1.5 CMCFG1CSn : OSPI Command Map Configuration Register 1 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x014 + 0x010 × n

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20–16 | 15–0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | RDLATE[4:0] | RDCMD[15:0] |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 1 0 0 0 | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 15:0 | RDCMD[15:0] | Read Command<br>These bits configure the command field of a read transaction in memory-mapping mode. Normal format and 8D-8D-8D profile 2.0 format use only upper 1 byte. 8D-8D-8D profile 1.0 format uses 2 bytes. | R/W |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 20:16 | RDLATE[4:0] | Read Latency Cycle<br>These bits configure the latency cycle of a read transaction in memory-mapping mode.<br>0x00: No latency<br>0x01: 1 cycle<br>⋮ ⋮<br>0x08: 8 cycles (default)<br>⋮ ⋮<br>0x1E: 30 cycles<br>0x1F: 31 cycles<br>Others: Setting prohibited | R/W |
| 31:21 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CMCFG1CSn register has functions to configure the xSPI Master function.

### 45.2.1.6 CMCFG2CSn : OSPI Command Map Configuration Register 2 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x018 + 0x010 × n

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20–16 | 15–0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | WRLATE[4:0] | WRCMD[15:0] |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 1 0 0 0 | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 15:0 | WRCMD[15:0] | Write Command<br>These bits configure the command field of a write transaction in memory-mapping mode. Normal format and 8D-8D-8D profile 2.0 format use only upper 1 byte. 8D-8D-8D profile 1.0 format uses 2 bytes. | R/W |
| 20:16 | WRLATE[4:0] | Write Latency Cycle<br>These bits configure the latency cycle of a write transaction in memory-mapping mode.<br>0x00: No latency<br>0x01: 1 cycle<br>⋮ ⋮<br>0x08: 8 cycles (default)<br>⋮ ⋮<br>0x1E: 30 cycles<br>0x1F: 31 cycles<br>Others: Setting prohibited | R/W |
| 31:21 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CMCFG2CSn register has functions to configure xSPI the Master function.

### 45.2.1.7 LIOCFGCSn : OSPI Link I/O Configuration Register CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x050 + 0x004 × n

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | DDRSMPEX[3:0] | | | | SDRSMPSFT[3:0] | | | | SDRSMPMD | SDRDRV | CSNEGEX | CSASTEX | CSMIN[3:0] | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | WRMSKMD | LATEMD | PRTMD[9:0] | | | | | | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 9:0 | PRTMD[9:0] | Protocol Mode<br>These bits configure the protocol mode and the pin to sample data inputs.<br>When using data strobe instead of SPI clock for sampling in SDR mode, set PRTMD[9:0] to 1.<br>0x000: 1S-1S-1S<br>0x3B2: 4S-4D-4D<br>0x3FF: 8D-8D-8D<br>0x048: 1S-2S-2S<br>0x049: 2S-2S-2S<br>0x090: 1S-4S-4S<br>0x092: 4S-4S-4S<br>Others: Setting prohibited | R/W |
| 10 | LATEMD | Latency Mode<br>This bit selects the behavior of the initial access latency phase for both direct-manual mode and memory-mapping mode. When set to 0, the latency cycle is equal to each configured cycle from the transmitting Address field. When set to 1, the latency cycle is incremented from the last byte pair of the Address field and is extended 2 times of each configured cycle depending on the OM_DQS port.<br>Latency mode is used only for profile 2.0 frame format of 8D-8D-8D protocol mode with 6 bytes Command/Address field. See the xSPI protocol specification for detail.<br><table><tr><th>Value: Function</th><th>Frame format</th><th>Usage</th></tr><tr><td rowspan="2">0: Configurable latency</td><td>Profile 2.0</td><td>The configurable latency cycle should be set as minus 1.</td></tr><tr><td>Others</td><td>The latency cycle increments after the Address field.</td></tr><tr><td rowspan="2">1: Variable latency</td><td>Profile 2.0</td><td>The latency cycle increments from address [23:16] and should not be set to 1.</td></tr><tr><td>Others</td><td>Not support</td></tr></table> | R/W |
| 11 | WRMSKMD | Write Mask Mode<br>This bit selects to use the OM_DQS port as the write data mask. It can be useful for write access of odd byte.<br>Write mask mode is used only for 8D-8D-8D protocol mode.<br>0: Write mask disable<br>1: Write mask enable | R/W |
| 15:12 | — | These bits are read as 0. The write value should be 0. | R/W |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 19:16 | CSMIN[3:0] | CS Minimum Idle Term<br>This bit configures the minimum cycle between xSPI frames.<br>0x0: 1 cycle<br>0x1: 2 cycles<br>⋮ ⋮<br>0x7: 8 cycles (default)<br>⋮ ⋮<br>0xE: 15 cycles<br>0xF: 16 cycles<br>Others: Setting prohibited | R/W |
| 20 | CSASTEX | CS Asserting Extension<br>This bit extends 1 cycle Chip Select pins when asserting.<br>0: No extension<br>1: Extend 1 cycle | R/W |
| 21 | CSNEGEX | CS Negating Extension<br>This bit extends 1 cycle Chip Select pins when negating.<br>0: No extension<br>1: Extend 1 cycle | R/W |
| 22 | SDRDRV | SDR Driving Timing<br>This bit configures the timing of data output in SDR.<br>This bit should not be set to 1 when there is no latency cycle because OSPI output data can be conflicted with OSPI input data.<br>0: Drive at 1/2 cycle before CK rising edge<br>1: Drive at CK rising-edge | R/W |
| 23 | SDRSMPMD | SDR Sampling Mode<br>This bit selects the edge of sampling in SDR. In DDR, regardless of the setting, this bit samples OM_SIOn (n = 0 to 7) ports with both edges of OM_DQS.<br>When set to 1, this bit samples at rising edge before falling edge.<br>0: Samples data input at falling edge<br>1: Samples data input at rising edge | R/W |
| 27:24 | SDRSMPSFT[3:0] | SDR Sampling window shift<br>These bits shift the timing of CK sampling in SDR. When using OM_DQS in SDR, there is no influence on the behavior.<br>For DDR or when using OM_DQS in SDR, set these bits to 0.<br>0x0: Sample without delay<br>0x1: Sample at 1-cycle delay<br>⋮ ⋮<br>0x6: Sample at 6-cycle delay<br>0x7: Sample at 7-cycle delay<br>Others: Setting prohibited | R/W |
| 31:28 | DDRSMPEX[3:0] | DDR Sampling Window Extend<br>These bits configure the cycle of extending the sampling window in DDR. In DDR, the input data is sampled during the expected cycle soon after the latency cycle. The input data out of range is ignored. It can be configured depending on the OM_DQS propagation delay.<br>These bits setting should be set to satisfy $t_{CKHDSH}$ in section 70.3.11. OSPI Timing. $t_{CKHDSH}$ means the output delay time of the external memory. For example, the DDRSMPEX[3:0] bits are set to 0x1 in the following conditions.<br>● VCC is 2.7 V or above<br>● OM_n_SCLK is 133 MHz<br>● Memory output delay time ($t_{CKHDSH}$) is 6 ns<br>The maximum allowable $t_{CKHDSH}$ value is: $t_{PERIOD}$ × (1 + DDRSMPEX[3:0]) - 8.5 = 7.5 × (1 + 1) - 8.5 = 6.5[ns]<br>The memory output delay time should be less than this value.<br>0x0: Expand no cycle<br>0x1: Expand 1 cycle<br>⋮ ⋮<br>0x6: Expand 6 cycles<br>0x7: Expand 7 cycles<br>Others: Setting prohibited | R/W |

Note: S-TYPE-3, P-TYPE-3

The LIOCFGCSn register has functions to configure the xSPI Master function.

## 45.2.2 OSPI Control Registers

These registers control the xSPI Master function.

### 45.2.2.1 BMCTL0 : OSPI Bridge Map Control Register 0

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x060

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | CH1CS1ACC[1:0] | | CH1CS0ACC[1:0] | | CH0CS1ACC[1:0] | | CH0CS0ACC[1:0] | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 1:0 | CH0CS0ACC[1:0] | System bus ch0 to slave 0 memory area access enable<br>This field enables the access from ch0 to CS0 memory.<br>0 0: Read/write disable<br>0 1: Read enable, write disable<br>1 0: Read disable, write enable<br>1 1: Read/write enable | R/W |
| 3:2 | CH0CS1ACC[1:0] | System bus ch0 to slave 1 memory area access enable<br>This field enables the access from ch0 to CS1 memory.<br>0 0: Read/write disable<br>0 1: Read enable, write disable<br>1 0: Read disable, write enable<br>1 1: Read/write enable | R/W |
| 5:4 | CH1CS0ACC[1:0] | System bus ch1 to slave 0 memory area access enable<br>This field enables the access from ch1 to CS0 memory.*1<br>0 0: Read/write disable<br>0 1: Read enable, write disable<br>1 0: Read disable, write enable<br>1 1: Read/write enable | R/W |
| 7:6 | CH1CS1ACC[1:0] | System bus ch1 to slave 1 memory area access enable<br>This field enables the access from ch1 to CS1 memory.*1<br>0 0: Read/write disable<br>0 1: Read enable, write disable<br>1 0: Read disable, write enable<br>1 1: Read/write enable | R/W |
| 31:8 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3
Note 1. ch1 can only be used for GLCDC1 bus master.

The BMCTL0 register has functions to control the xSPI Master function.

This register is configured in the initialization phase. When the setting requires modifications after the start of an xSPI transaction, stop all communication. See section 45.3.7.2. Flow of Communication Stop before changing the value of BMCTL0.

### 45.2.2.2 BMCTL1 : OSPI Bridge Map Control Register 1

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x064

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | PBUFCLRCH1 | PBUFCLRCH0 | MWRPUSHCH1 | MWRPUSHCH0 | — | — | — | — | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 7:0 | — | The write value should be 0. | W |
| 8 | MWRPUSHCH0 | Memory Write Data Push for ch0<br>This field requests to push the pending data in combination mode.*1<br>0: No command<br>1: Push request | W |
| 9 | MWRPUSHCH1 | Memory Write Data Push for ch1*2<br>The function is the same as one of ch0.*1 | W |
| 10 | PBUFCLRCH0 | Prefetch Buffer Clear for ch0<br>This field requests to clear the prefetch buffer when the prefetch function is enabled.*1<br>Do not set this bit during memory access (COMSTT.MEMACCCH0 = 1).<br>0: No command<br>1: Clear request | W |
| 11 | PBUFCLRCH1 | Prefetch Buffer Clear for ch1*2<br>This function is the same as one of ch0.*1 | W |
| 31:12 | — | The write value should be 0. | W |

Note: S-TYPE-3, P-TYPE-3
Note 1. Prefetch/Combination behavior is not defined when the cycle base race condition between asserting these bits and system bus access has occurred.
Note 2. ch1 can only be used for GLCDC1 bus master.

The BMCTL1 register has functions to control the xSPI Master function.

### 45.2.2.3 CMCTLCHn : OSPI Command Map Control Register CHn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x0068 + 0x004 × n (n = 0, 1)

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | XIPEN |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | XIPEXCODE[7:0] | | | | | | | | XIPENCODE[7:0] | | | | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 7:0 | XIPENCODE[7:0] | XiP Mode Enter Code<br>These bits configure the code to enter XiP mode in memory-mapping mode. | R/W |
| 15:8 | XIPEXCODE[7:0] | XiP Mode Exit Code<br>These bits configure the code to exit XiP mode in memory-mapping mode. | R/W |
| 16 | XIPEN | XiP Mode Enable<br>This bit enables XiP mode in memory-mapping mode. When set to 1, XiP enter code is inserted in the latency field, and the command field in the next transaction is omitted. When set to 0, XiP exit code is inserted in the latency field. This bit is set to 0 automatically when transmitting a XiP disable pattern.<br>XiP mode should not be used for 8D-8D-8D protocol mode profile 2.0 frame format.<br>0: Disable XiP mode<br>1: Enable XiP mode | R/W |
| 31:17 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CMCTLCHn register has functions to control the xSPI Master function.

### 45.2.2.4 CDCTL0 : OSPI Command Manual Control Register 0

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x070

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | PERREP[3:0] | | | | — | — | — | PERITV[4:0] | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | TRNUM[1:0] | | CSSEL | — | PERMD | TRREQ |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | TRREQ | Transaction Request<br>This bit requests to issue the transaction of manual-command. When set to 1, it starts the transaction. This bit is cleared to 0 when the transaction is completed.<br>The transaction is canceled by clearing to 0 while the transaction is ongoing.<br>0: No transaction<br>1: Request transaction | R/W |
| 1 | PERMD | Periodic Mode<br>This bit enables the periodic transaction mode. When set to 1, it repeats a transaction periodically and compares the read value with the expected value. Periodic transaction mode alternates the status polling operation for external memory.<br>0: Direct manual-command mode<br>1: Periodic manual-command mode | R/W |
| 2 | — | This bit is read as 0. The write value should be 0. | R/W |
| 3 | CSSEL | Chip Select<br>This bit selects a target memory to issue a manual-command.<br>0: CS0<br>1: CS1 | R/W |
| 5:4 | TRNUM[1:0] | Transaction Number<br>These bits configure the number of transactions in normal manual-command mode. In periodic manual-command, regardless of this setting, the read data of the last command is compared.<br>0 0: Issue 1 command (using command buffer 0)<br>0 1: Issue 2 commands (using command buffer 0-1)<br>1 0: Issue 3 commands (using command buffer 0-2)<br>1 1: Issue 4 commands (using command buffer 0-3) | R/W |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 15:6 | — | These bits are read as 0. The write value should be 0. | R/W |
| 20:16 | PERITV[4:0] | Periodic Transaction Interval<br>These bits configure the interval of transaction in periodic manual-command mode.<br>Too short interval compared with CPU bus cycle can result in no store into command buffer0. The interval should be longer than 4 times the CPU bus cycle.<br>0x00: 2 (= $2^1$) cycles<br>0x01: 4 (= $2^2$) cycles<br>⋮ ⋮<br>0x1E: 2,147,483,648 (= $2^{31}$) cycles<br>0x1F: 4,294,967,296 (= $2^{32}$) cycles | R/W |
| 23:21 | — | These bits are read as 0. The write value should be 0. | R/W |
| 27:24 | PERREP[3:0] | Periodic Transaction Repeat<br>These bits configure the number of transaction repetitions in periodic manual-command mode.<br>0x0: 1 (= $2^0$) time<br>0x1: 2 (= $2^1$) times<br>⋮ ⋮<br>0xE: 16384 (= $2^{14}$) times<br>0xF: 32768 (= $2^{15}$) times | R/W |
| 31:28 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CDCTL0 register has functions to control the xSPI Master function.

### 45.2.2.5 CDCTL1 : OSPI Command Manual Control Register 1

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x074

| Bit position: | 31 ... 0 |
|---|---|
| Bit field: | PEREXP[31:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | PEREXP[31:0] | Periodic Transaction Expected Value<br>These bits configure the expected value to compare with the read value in periodic manual-command mode. For example, when comparing one byte, the lower byte should be configured. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CDCTL1 register has functions to control the xSPI Master function.

### 45.2.2.6 CDCTL2 : OSPI Command Manual Control Register 2

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x078

| Bit position: | 31 ... 0 |
|---|---|
| Bit field: | PERMSK[31:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | PERMSK[31:0] | Periodic Transaction Masked Value<br>These bits configure the masked value for the expected value in periodic manual-command mode.<br>When 1 is set to any bit, the corresponding bit is configured as the expected value (CDCTL1.PEREXP[31:0]) is ignored.<br>In 8D-8D-8D, the data bytes are transferred only in byte pairs on the xSPI bus. This means the dummy read data can be stored. It should be masked for unused bits. For example, when reading a lower one byte, it should be configured to 0xFFFFFF00. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CDCTL2 register has functions to control the xSPI Master function.

### 45.2.2.7 CDTBUFn : OSPI Command Manual Type Buf n (n = 0 to 3)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x080 + 0x010 × n

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | CMD[15:0] | | | | | | | | | | | | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | TRTYPE | — | LATE[4:0] | | | | | DATASIZE[3:0] | | | | ADDSIZE[2:0] | | | CMDSIZE[1:0] | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 1:0 | CMDSIZE[1:0] | Command Size<br>These bits configure the size of the Command field.<br>In 8D-8D-8D, set these bits to 10b.<br>Do not configure both command size and address size to 0.<br>0 0: 0 bytes (no command phase)<br>0 1: 1 byte<br>1 0: 2 bytes<br>Others: Setting prohibited | R/W |
| 4:2 | ADDSIZE[2:0] | Address Size<br>These bits configure the size of the address field.<br>0 0 0: 0 bytes (no address phase)<br>0 0 1: 1 byte<br>0 1 0: 2 bytes<br>0 1 1: 3 bytes<br>1 0 0: 4 bytes<br>Others: Setting prohibited | R/W |
| 8:5 | DATASIZE[3:0] | Write/Read Data Size<br>These bits configure the size of the Data field.<br>In 8D-8D-8D, the data bytes are transferred only in byte pairs on the xSPI bus. For example, even when configuring 1 byte for read, 2 bytes of data are received. The last byte should be ignored.<br>Do not configure 0 bytes for read transaction.<br>0x0: 0 bytes (no data phase)<br>0x1: 1 byte<br>⋮ ⋮<br>0x7: 7 bytes<br>0x8: 8 bytes<br>Others: Setting prohibited | R/W |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 13:9 | LATE[4:0] | Latency Cycle<br>These bits configure the latency cycle in manual-command mode.<br>0x0: No latency<br>0x1: 1 cycle<br>⋮ ⋮<br>0x1E: 30 cycles<br>0x1F: 31 cycles | R/W |
| 14 | — | This bit is read as 0. The write value should be 0. | R/W |
| 15 | TRTYPE | Transaction Type<br>This bit selects the type of transaction.<br>0: Read transaction (readout data from slave device)<br>1: Not read transaction | R/W |
| 31:16 | CMD[15:0] | Command (1-2 bytes)<br>These bits configure the Command field in manual-command mode.<br>The number of bytes configured in Command Size bit is transferred.<br>1S-1S-1S, 4S-4D-4D:<br>CMD[15:8] is Command field, CMD[7:0] is not used.<br>8D-8D-8D profile 1.0:<br>CMD[15:8] is Command field, CMD[7:0] is Extension field.<br>8D-8D-8D profile 2.0:<br>CMD[15:0] is the upper 2 bytes of the Command & Modifier fields (bits [47:32] in the xSPI protocol) | R/W |

Note: S-TYPE-3, P-TYPE-3

The CDTBUFn register has functions to control the xSPI Master function.

### 45.2.2.8 CDABUFn : OSPI Command Manual Address Buf n (n = 0 to 3)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x084 + 0x10 × n

| | |
|---|---|
| Bit position: | 31 … 0 |
| Bit field: | ADD[31:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | ADD[31:0] | Address<br>These bits configure the Address field in manual-command mode.<br>1S-1S-1S, 4S-4D-4D, 8D-8D-8D profile 1.0: The Address field.<br>8D-8D-8D profile 2.0: The lower 4 bytes of the Command & Modifier fields (bits [31:0] in the xSPI protocol). | R/W |

Note: S-TYPE-3, P-TYPE-3

The CDABUFn register has functions to control the xSPI Master function.

### 45.2.2.9 CDD0BUFn : OSPI Command Manual Data 0 Buf n (n = 0 to 3)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x088 + 0x10 × n

| | |
|---|---|
| Bit position: | 31 … 0 |
| Bit field: | DATA[31:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | DATA[31:0] | Write/Read Data<br>These bits configure the Data field in manual-command mode.<br>For write transaction, configure the write data.<br>For read transaction, the read data is stored. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CDD0BUFn register has functions to control the xSPI Master function.

### 45.2.2.10 CDD1BUFn : OSPI Command Manual Data 1 Buf n (n = 0 to 3)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x08C + 0x10 × n

| Bit position: | 31 … 0 |
|---|---|
| Bit field: | DATA[31:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | DATA[31:0] | Write/Read Data<br>These bits configure the Data field in manual-command mode.<br>For write transaction, configure the write data.<br>For read transaction, the read data is stored. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CDD1BUFn register has functions to control the xSPI Master function.

### 45.2.2.11 LPCTL0 : OSPI Link Pattern Control Register 0

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x100

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | XD2VAL | — | — | XD2LEN[4:0] | | | | | XD1VAL | — | — | XD1LEN[4:0] | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | XDPIN[1:0] | | CSSEL | — | — | PATREQ |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | PATREQ | Pattern Request<br>This bit requests to issue the pattern. When set to 1, it starts the pattern. It is cleared to 0 when the pattern is completed.<br>0: No request XiP disable pattern<br>1: Request XiP disable pattern | R/W |
| 2:1 | — | These bits are read as 0. The write value should be 0. | R/W |
| 3 | CSSEL | Chip Select<br>This bit selects a target memory to issue a pattern.<br>0: Slave 0 (CS0)<br>1: Slave 1 (CS1) | R/W |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 5:4 | XDPIN[1:0] | XiP Disable Pattern Pin<br>These bits select the data output pins to transmit XiP disable pattern.<br>0 0: 1 pin<br>0 1: 2 pins<br>1 0: 4 pins<br>1 1: 8 pins | R/W |
| 15:6 | — | These bits are read as 0. The write value should be 0. | R/W |
| 20:16 | XD1LEN[4:0] | XiP Disable Pattern 1st Phase Length<br>These bits select the length of 1st phase in XiP disable pattern.<br>Do not configure the pattern with zero-length for both 1st phase and 2nd phase.<br>0x0: 0 cycles<br>0x1: 1 cycle<br>⋮ ⋮<br>0x1E: 30 cycles<br>0x1F: 31 cycles | R/W |
| 22:21 | — | These bits are read as 0. The write value should be 0. | R/W |
| 23 | XD1VAL | XiP Disable Pattern 1st Phase Value<br>This bit selects the value of 1st phase in XiP disable pattern.<br>0: Low drive<br>1: High drive | R/W |
| 28:24 | XD2LEN[4:0] | XiP Disable Pattern 2nd Phase Length<br>These bits select the length of 2nd phase in XiP disable pattern.<br>0x00: 0 cycles<br>0x01: 1 cycle<br>⋮ ⋮<br>0x1E: 30 cycles<br>0x1F: 31 cycles | R/W |
| 30:29 | — | These bits are read as 0. The write value should be 0. | R/W |
| 31 | XD2VAL | XiP Disable Pattern 2nd Phase Value<br>This bit selects the value of 2nd phase in XiP disable pattern.<br>0: Low drive<br>1: High drive | R/W |

Note: S-TYPE-3, P-TYPE-3

The LPCTL0 register has functions to control the xSPI Master function.

### 45.2.2.12 LPCTL1 : OSPI Link Pattern Control Register 1

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x104

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | RSTSU[2:0] | | | — | RSTWID[2:0] | | | — | — | RSTREP[1:0] | | CSSEL | — | PATREQ[1:0] | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 1:0 | PATREQ[1:0] | Pattern Request<br>These bits request to issue the pattern. When set to 01b or 10b, it starts the pattern. It is cleared to 00b when the pattern is completed.<br>0 0: No request<br>0 1: Request Reset pattern<br>1 0: Request CS-only pattern<br>1 1: Setting prohibited | R/W |
| 2 | — | This bit is read as 0. The write value should be 0. | R/W |
| 3 | CSSEL | Chip select<br>This bit selects a target memory to issue a pattern.<br>0: Slave 0 (CS0)<br>1: Slave 1 (CS1) | R/W |
| 5:4 | RSTREP[1:0] | Reset Pattern Repeat<br>These bits select the repeating time to toggle CS from low to high.<br>0 0: 4 times (specified on Reset Signaling Protocol)<br>0 1: 5 times<br>1 0: 6 times<br>1 1: 7 times | R/W |
| 7:6 | — | These bits are read as 0. The write value should be 0. | R/W |
| 10:8 | RSTWID[2:0] | Reset Pattern Width<br>These bits configure the width of a cycle in Reset pattern and CS-only pattern. It toggles CS with the configured cycle.<br>0 0 0: 2 (= $2^1$) cycles<br>0 0 1: 4 (= $2^2$) cycles<br>⋮ ⋮<br>1 1 0: 128 (= $2^7$) cycles<br>1 1 1: 256 (= $2^8$) cycles | R/W |
| 11 | — | This bit is read as 0. The write value should be 0. | R/W |
| 14:12 | RSTSU[2:0] | Reset Pattern Data Output Setup Time<br>These bits configure the number of setup cycles for data output based on the edge of CS in a Reset pattern. The number of setup cycles requires enough setup time because the xSPI slave samples any data at the rising edge of CS.<br>This cycle of setup time should be less than the cycle of reset pattern width (RSTWID[2:0]).<br>0 0 0: 1 cycle<br>0 0 1: 2 cycles<br>⋮ ⋮<br>1 1 0: 7 cycles<br>1 1 1: 8 cycles | R/W |
| 31:15 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

The LPCTL1 register has functions to control the xSPI Master function.

### 45.2.2.13 LIOCTL : OSPI Link I/O Control Register

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x108

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | RSTCS0 |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | WPCS1 | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | — | This bit is read as 1. The write value should be 1. | R/W |
| 1 | WPCS1 | WP Drive for Slave 1<br>This bit controls the value of the OM_WP1 port. It can be useful only for xSPI slave with write protect port.<br>0: Drive low level<br>1: Drive high level | R/W |
| 15:2 | — | These bits are read as 0. The write value should be 0. | R/W |
| 16 | RSTCS0 | Reset Drive<br>This bit controls the value of OM_RESET port. It can be useful only for xSPI slave with reset port.<br>0: Drive low level<br>1: Drive high level | R/W |
| 17 | — | This bit is read as 1. The write value should be 1. | R/W |
| 31:18 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

The LIOCTL register has functions to control the xSPI Master function.

### 45.2.2.14 CCCTL0CSn : OSPI Command Calibration Control Register 0 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x130 + 0x020 × n

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | CASFTEND[4:0] | | | | | — | — | — | CASFTSTA[4:0] | | | | |
| Value after reset: | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | CAITV[4:0] | | | | | — | — | — | — | — | — | CANOWR | CAEN |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | CAEN | Automatic Calibration Enable<br>This bit enables the automatic calibration. When set to 1, it transmits the calibration sequence periodically and adjusts the value of phase shift. When set to 0 during the calibration sequence, it stops after completion of the ongoing calibration sequence, this bit is then cleared.<br>0: Disable automatic calibration<br>1: Enable automatic calibration | R/W |
| 1 | CANOWR | Calibration No Write Mode<br>This bit selects to omit write command in the calibration sequence. It can be used for any slave device with fixed calibration pattern data.<br>0: Calibration sequence with write command<br>1: Calibration sequence without write command | R/W |
| 7:2 | — | These bits are read as 0. The write value should be 0. | R/W |
| 12:8 | CAITV[4:0] | Calibration Interval<br>These bits configure the intervals between calibration patterns.<br>0x00: 2 (= $2^1$) cycle wait<br>0x01: 4 (= $2^2$) cycle wait<br>⋮<br>0x1E: 2,147,483,648 (= $2^{31}$) cycle wait<br>0x1F: 4,294,967,296 (= $2^{32}$) cycle wait | R/W |
| 15:13 | — | These bits are read as 0. The write value should be 0. | R/W |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 20:16 | CASFTSTA[4:0] | Calibration DS Shift Start Value<br>These bits configure the start value of DS shift. | R/W |
| 23:21 | — | These bits are read as 0. The write value should be 0. | R/W |
| 28:24 | CASFTEND[4:0] | Calibration DS Shift End Value<br>These bits configure the end value of DS shift.<br>The end value should be equal to or more than the start value (CASFTSTA[4:0]). | R/W |
| 31:29 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CCCTL0CSn register has functions to control the xSPI Master function.

### 45.2.2.15 CCCTL1CSn : OSPI Command Calibration Control Register 1 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x134 + 0x020 × n

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | CARDLATE[4:0] | | | | | — | — | — | CAWRLATE[4:0] | | | | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | CADATASIZE[3:0] | | | | CAADDSIZE[2:0] | | | CACMDSIZE [1:0] | |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 1:0 | CACMDSIZE[1:0] | Command Size<br>These bits configure the size of the Command field.<br>In 8D-8D-8D, set these bits to 10b.<br>Do not configure both command size and address size to 0.<br>0 0: 0 bytes (no command phase)<br>0 1: 1 byte<br>1 0: 2 bytes<br>1 1: Setting prohibited | R/W |
| 4:2 | CAADDSIZE[2:0] | Address Size<br>These bits configure the size of the Address field.<br>0 0 0: 0 bytes (no address phase)<br>0 0 1: 1 byte<br>0 1 0: 2 bytes<br>0 1 1: 3 bytes<br>1 0 0: 4 bytes<br>Others: Setting prohibited | R/W |
| 8:5 | CADATASIZE[3:0] | Write/Read Data Size<br>These bits configure the size of the Data field.<br>In 8D-8D-8D, the data size should be configured with an even byte.<br>0x0: 1 byte<br>0x1: 2 bytes<br>⋮ ⋮<br>0xE: 15 bytes<br>0xF: 16 bytes | R/W |
| 15:9 | — | These bits are read as 0. The write value should be 0. | R/W |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 20:16 | CAWRLATE[4:0] | Write Latency Cycle<br>These bits configure the latency cycle in the calibration frame.<br>0x00: No latency<br>0x01: 1 cycle<br>⋮ ⋮<br>0x1E: 30 cycles<br>0x1F: 31 cycles | R/W |
| 23:21 | — | These bits are read as 0. The write value should be 0. | R/W |
| 28:24 | CARDLATE[4:0] | Read Latency Cycle<br>These bits configure the latency cycle in the calibration frame.<br>0x00: No latency<br>0x01: 1 cycle<br>⋮ ⋮<br>0x1E: 30 cycles<br>0x1F: 31 cycles | R/W |
| 31:29 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CCCTL1CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

### 45.2.2.16 CCCTL2CSn : OSPI Command Calibration Control Register 2 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x138 + 0x020 × n

| Bit position: | 31–16 | 15–0 |
|---|---|---|
| Bit field: | CARDCMD[15:0] | CAWRCMD[15:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 15:0 | CAWRCMD[15:0] | Calibration Pattern Write Command<br>These bits configure the calibration pattern write command. | R/W |
| 31:16 | CARDCMD[15:0] | Calibration Pattern Read Command<br>These bits configure the calibration pattern read command. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CCCTL2CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

### 45.2.2.17 CCCTL3CSn : OSPI Command Calibration Control Register 3 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x13C + 0x020 × n

| Bit position: | 31–0 |
|---|---|
| Bit field: | CAADD[31:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | CAADD[31:0] | Calibration Pattern Address<br>These bits configure the calibration pattern address. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CCCTL3CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

### 45.2.2.18 CCCTL4CSn : OSPI Command Calibration Control Register 4 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x140 + 0x020 × n

| Bit position: | 31 … 0 |
|---|---|
| Bit field: | CADATA[31:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | CADATA[31:0] | Calibration Pattern Data<br>These bits configure the calibration pattern data. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CCCTL4CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

### 45.2.2.19 CCCTL5CSn : OSPI Command Calibration Control Register 5 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x144 + 0x020 × n

| Bit position: | 31 … 0 |
|---|---|
| Bit field: | CADATA[31:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | CADATA[31:0] | Calibration Pattern Data<br>These bits configure the calibration pattern data. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CCCTL5CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

### 45.2.2.20 CCCTL6CSn : OSPI Command Calibration Control Register 6 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x148 + 0x020 × n

| Bit position: | 31 … 0 |
|---|---|
| Bit field: | CADATA[31:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | CADATA[31:0] | Calibration Pattern Data<br>These bits configure the calibration pattern data. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CCCTL6CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

### 45.2.2.21 CCCTL7CSn : OSPI Command Calibration Control Register 7 CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x14C + 0x020 × n

| Bit position: | 31 … 0 |
|---|---|
| Bit field: | CADATA[31:0] |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | CADATA[31:0] | Calibration Pattern Data<br>These bits configure the calibration pattern data. | R/W |

Note: S-TYPE-3, P-TYPE-3

The CCCTL7CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

### 45.2.2.22 RACTL : OSPI Register Access Control Register

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x208

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | RAEN |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | RAEN | Register Access Enable<br>0: Disable MER register access<br>1: Enable MER register access | R/W |
| 31:1 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

### 45.2.2.23 MER : OSPI Map End Replace Register

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x228

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | MECS1 | | | | | | | | — | — | — | MERCS1 |
| Value after reset: | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 3:0 | — | These bits are read as 0. The write value should be 0. | R/W |
| 11:4 | — | These bits are read as 1. The write value should be 1. | R/W |
| 15:12 | — | These bits are read as 0. The write value should be 0. | R/W |
| 16 | MERCS1 | Map End Replace CS1 Enable<br>0: No replacement<br>1: Replacement | R/W |
| 19:17 | — | These bits are read as 0. The write value should be 0. | R/W |
| 27:20 | MECS1 | Map End CS1 Replace Address<br>The bit27 to bit20 of system bus address is replaced with this value when MERCS1 is set to 1. | R/W |
| 31:28 | — | These bits are read as 0. The write value should be 0. | R/W |

Note: S-TYPE-3, P-TYPE-3

For SiP product, the MER register should be set as described in section 45.4. SiP Product Configuration. For standard product, do not change the value after reset.

## 45.2.3 OSPI Status Registers

These registers monitor the status of xSPI Master.

### 45.2.3.1 COMSTT : OSPI Common Status Register

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x184

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | — | RSTOCS1 | INTCS1 | ECSCS1 | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | — | — | — | — | — | — | WRBUFNECH1 | WRBUFNECH0 | PBUFNECH1 | PBUFNECH0 | — | — | MEMACCCH1 | MEMACCCH0 |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | MEMACCCH0 | Memory Access Ongoing from Channel 0<br>This bit is only valid in the section 45.3.7.2. Flow of Communication Stop and section 45.3.7.6. Flow of Memory-mapping Stop flows.<br>0: System bus bridge CH0 is not accessing to memory<br>1: System bus bridge CH0 is accessing to memory | R |
| 1 | MEMACCCH1 | Memory Access Ongoing from Channel 1*1<br>This function is the same as CH0. | R |
| 3:2 | — | These bits are read as 0. | R |
| 4 | PBUFNECH0 | Prefetch Buffer Not Empty for Channel 0<br>0: Empty<br>1: Not empty | R |
| 5 | PBUFNECH1 | Prefetch Buffer Not Empty for Channel 1*1<br>This function is the same as CH0. | R |
| 6 | WRBUFNECH0 | Write Buffer Not Empty for Channel 0<br>This bit is only valid in the section 45.3.7.2. Flow of Communication Stop and section 45.3.7.6. Flow of Memory-mapping Stop flows.<br>0: Empty<br>1: Not empty | R |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 7 | WRBUFNECH1 | Write Buffer Not Empty for Channel 1*1<br>This function is the same as CH0. | R |
| 19:8 | — | These bits are read as 0. | R |
| 20 | ECSCS1 | ECS Monitor for Slave 1<br>This bit indicates the value of OM_ECSINT1 port.<br>0: Low level<br>1: High level | R |
| 21 | INTCS1 | INT Monitor for Slave 1<br>This bit indicates the value of OM_ECSINT1 port.<br>0: Low level<br>1: High level | R |
| 22 | RSTOCS1 | RSTO Monitor for Slave 1<br>This bit indicates the value of OM_RSTO1 port.<br>0: Low level<br>1: High level | R |
| 31:23 | — | These bits are read as 0. | R |

Note: S-TYPE-3, P-TYPE-3
Note 1. ch1 can only be used for GLCDC1 bus master.

The COMSTT register indicates the status of the xSPI Master.

#### 45.2.3.2 CASTTCSn : OSPI Calibration Status Register CSn (n = 0, 1)

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x188 + 0x004 × n

| Bit position: | 31 | 0 |
|---|---|---|
| Bit field: | CASUC[31:0] | |
| Value after reset: | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 | |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 31:0 | CASUC[31:0] | Calibration Success<br>These bits indicate the calibration success for each DS shift value.<br>This bit is updated for each completed calibration sequence.<br>CASUC[x] indicates calibration success in DS shift value = x. | R |

Note: S-TYPE-3, P-TYPE-3

The CASTTCSn register indicates the status of xSPI Master.

### 45.2.4 OSPI Interrupt Registers

These registers control the interrupt function of xSPI Master.

### 45.2.4.1 INTS : OSPI Interrupt Status Register

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x190

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | CASUCCS1 | CASUCCS0 | CAFAILCS1 | CAFAILCS0 | — | — | — | — | — | — | BUSERRCH1 | BUSERRCH0 | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | INTCS1 | — | — | — | ECSCS1 | — | — | — | DSTOCS1 | DSTOCS0 | PERTO | — | PATCMP | CMDCMP |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | CMDCMP | Command Completed<br>This bit is set to 1 when the requested manual-command is completed.<br>In direct manual-command, it means all transactions completed.<br>In periodic manual-command, it means the read data matches with the expected data.<br>0: No detection<br>1: Detection | R |
| 1 | PATCMP | Pattern Completed<br>This bit is set to 1 when the requested pattern is completed.<br>0: No detection<br>1: Detection | R |
| 2 | — | This bit is read as 0. | R |
| 3 | PERTO | Periodic Transaction Timeout<br>This bit is set to 1 when the read value does not match with the expected value in periodic manual-command mode.<br>0: No detection<br>1: Detection | R |
| 4 | DSTOCS0 | DS Timeout for Slave 0<br>This bit is set to 1 for lost DS in read transaction when using DS.<br>It means not receiving the data during the expected read phase. In this case, xSPI master stops the read transaction and the following transaction.<br>This error may issue an error response to the system bus.<br>0: No detection<br>1: Detection | R |
| 5 | DSTOCS1 | DS Timeout for Slave 1<br>This function is the same as DSTOCS0.<br>0: No detection<br>1: Detection | R |
| 8:6 | — | These bits are read as 0. | R |
| 9 | ECSCS1 | ECC Error Detection for Slave 1<br>This bit is set to 1 when the falling edge is detected on the OM_ECSINT1 port. It can be useful only for xSPI slave with ECC detection function.<br>0: No detection<br>1: Detection | R |
| 12:10 | — | These bits are read as 0. | R |
| 13 | INTCS1 | Interrupt Detection for Slave 1<br>This bit is set to 1 when the falling edge is detected on the OM_ECSINT1 port. It can be useful only for xSPI slave with interrupt function.<br>0: No detection<br>1: Detection | R |
| 19:14 | — | These bits are read as 0. | R |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 20 | BUSERRCH0 | System Bus Error for Channel 0<br>This bit is set to 1 when an error response occurs on system bus CH0. | R |
| 21 | BUSERRCH1 | System Bus Error for Channel 1*1<br>This function is the same as CH0. | R |
| 27:22 | — | These bits are read as 0. | R |
| 28 | CAFAILCS0 | Calibration Failed for Slave 0<br>This bit is set to 1 when calibration failed.<br>0: No detection<br>1: Detection | R |
| 29 | CAFAILCS1 | Calibration Failed for Slave 1<br>This function is the same as CAFAILCS0.<br>0: No detection<br>1: Detection | R |
| 30 | CASUCCS0 | Calibration Success for Slave 0<br>This bit is set to 1 for success calibration.<br>0: No detection<br>1: Detection | R |
| 31 | CASUCCS1 | Calibration Success for Slave 1<br>This function is the same as CASUCCS0.<br>0: No detection<br>1: Detection | R |

Note: S-TYPE-3, P-TYPE-3
Note 1. ch1 can only be used for GLCDC1 bus master.

The INTS register indicates the status of an interrupt. The bits in this register are cleared to 0 when writing 1 on the corresponding bit of the INTC register.

### 45.2.4.2 INTC : OSPI Interrupt Clear Register

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x194

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | CASUCCS1C | CASUCCS0C | CAFAILCS1C | CAFAILCS0C | — | — | — | — | — | — | BUSERRCH1C | BUSERRCH0C | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | INTCS1C | — | — | — | ECSCS1C | — | — | — | DSTOCS1C | DSTOCS0C | PERTOC | — | PATCMPC | CMDCMPC |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | CMDCMPC | Command Completed Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 1 | PATCMPC | Pattern Completed Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 2 | — | The write value should be 0. | W |
| 3 | PERTOC | Periodic Transaction Timeout Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 4 | DSTOCS0C | DS Timeout for Slave 0 Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 5 | DSTOCS1C | DS Timeout for Slave 1 Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 8:6 | — | The write value should be 0. | W |
| 9 | ECSCS1C | ECC Error Detection for Slave 1 Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 12:10 | — | The write value should be 0. | W |
| 13 | INTCS1C | Interrupt Detection for Slave 1 Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 19:14 | — | The write value should be 0. | W |
| 20 | BUSERRCH0C | System Bus Error for ch0 Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 21 | BUSERRCH1C | System Bus Error for ch1 Interrupt Clear*1<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 27:22 | — | The write value should be 0. | W |
| 28 | CAFAILCS0C | Calibration Failed for Slave 0 Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 29 | CAFAILCS1C | Calibration Failed for Slave 1 Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 30 | CASUCCS0C | Calibration Success for Slave 0 Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |
| 31 | CASUCCS1C | Calibration Success for Slave 1 Interrupt Clear<br>0: No change interrupt status<br>1: Clear interrupt status | W |

Note: S-TYPE-3, P-TYPE-3

Note 1. ch1 can only be used for GLCDC1 bus master.

The INTC register clears the status of interrupt.

### 45.2.4.3 INTE : OSPI Interrupt Enable Register

Base address: OSPIn_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPIn_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x198

| Bit position: | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | CASUCCS1E | CASUCCS0E | CAFAILCS1E | CAFAILCS0E | — | — | — | — | — | — | BUSERRCH1E | BUSERRCH0E | — | — | — | — |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit position: | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit field: | — | — | INTCS1E | — | — | — | ECSCS1E | — | — | — | DSTOCS1E | DSTOCS0E | PERTOE | — | PATCMPE | CMDCMPE |
| Value after reset: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | Symbol | Function | R/W |
|---|---|---|---|
| 0 | CMDCMPE | Command Completed Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 1 | PATCMPE | Pattern Completed Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 2 | — | This bit is read as 0. The write value should be 0. | R/W |
| 3 | PERTOE | Periodic Transaction Timeout Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 4 | DSTOCS0E | DS Timeout for Slave 0 Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 5 | DSTOCS1E | DS Timeout for Slave 1 Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 8:6 | — | These bits are read as 0. The write value should be 0. | R/W |
| 9 | ECSCS1E | ECC Error Detection for Slave 1 Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 12:10 | — | These bits are read as 0. The write value should be 0. | R/W |
| 13 | INTCS1E | Interrupt Detection for Slave 1 Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 19:14 | — | These bits are read as 0. The write value should be 0. | R/W |
| 20 | BUSERRCH0E | System Bus Error for ch0 Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 21 | BUSERRCH1E | System Bus Error for ch1 Interrupt Enable*1<br>0: Disabled<br>1: Enabled | R/W |
| 27:22 | — | These bits are read as 0. The write value should be 0. | R/W |
| 28 | CAFAILCS0E | Calibration Failed for Slave 0 Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 29 | CAFAILCS1E | Calibration Failed for Slave 1 Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 30 | CASUCCS0E | Calibration Success for Slave 0 Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |
| 31 | CASUCCS1E | Calibration Success for Slave 1 Interrupt Enable<br>0: Disabled<br>1: Enabled | R/W |

Note: S-TYPE-3, P-TYPE-3
Note 1. ch1 can only be used for GLCDC1 bus master.

The INTE register enables the interrupt.

## 45.3 Operation

xSPI Master interface has the functions to issue the transaction for external memory with xSPI Slave interface. It allows writes to registers in external memory or reads from it.

This xSPI Master has two modes to issue transaction.

- Manual-command mode

- Memory-mapping mode

In manual-command mode, software configures all fields of xSPI frame and starts the transaction by a software request. In memory-mapping mode, software automatically converts a system bus for a preconfigured memory area into an xSPI transaction. It enables access from the system bus to the external memory area outside of the chip through xSPI bus.

This section describes the xSPI bus operation, the direct control of xSPI frame (manual-command), the control of memory access (memory-mapping), the error operation, and the flow to operation.

## 45.3.1 xSPI Bus

This section describes the xSPI bus operation.

Figure 45.2 shows an example of connections between OSPI and memory devices.

In this case, connect OM_CS0 to the RAM device and connect OM_CS1 to the Flash device.

Specify pull-up resistors according to the instructions for each device.

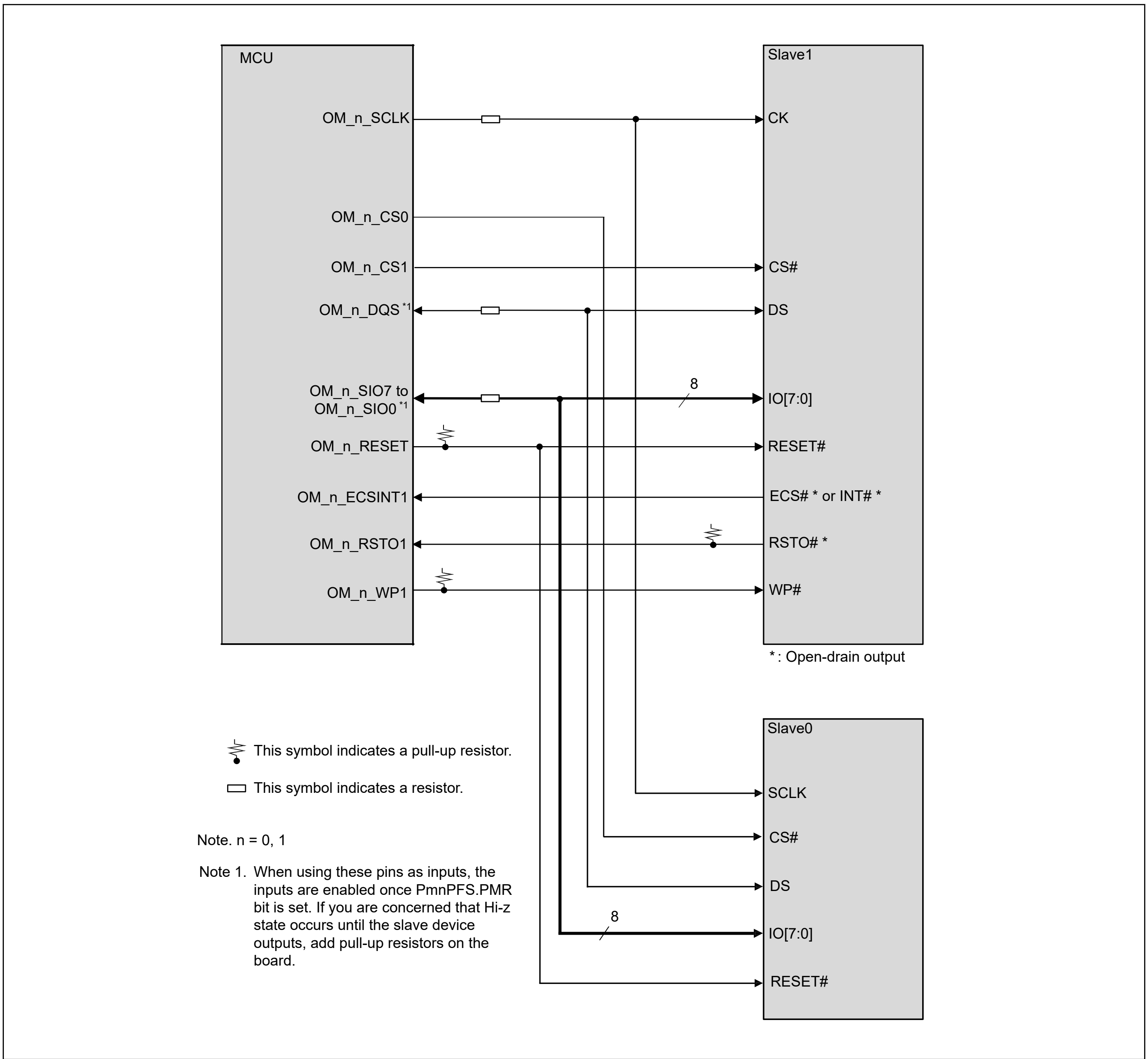


**Figure 45.2 Example of connection between OSPI and memory devices**

### 45.3.1.1 Supported Protocol Mode

This xSPI Master supports various protocol modes. It is configured by Protocol mode bits (LIOCFGCSn.PRTMD[9:0]). Table 45.3 shows the summary of protocol modes.

**Table 45.3 Supported protocol mode**

| Protocol mode | Function | PRTMD[9:0] | Note |
|---|---|---|---|
| 1S-1S-1S | Command, Address and Data fields are transferred at SDR using 1 data input pin and 1 data output pin. Read data is sampled with CK. | 0x000 | Specified by xSPI protocol |
| 4S-4D-4D | Command field is transferred at SDR using 4 data pins. Address and Data fields are transferred at DDR using 4 data pins. Read data is sampled with OM_DQS. | 0x3B2 | Specified by xSPI protocol |
| 8D-8D-8D | Command, Address and Data fields are transferred at DDR using 8 data pins. Read data is sampled with OM_DQS. | 0x3FF | Specified by xSPI protocol |
| 1S-2S-2S | Command field is transferred at SDR using 1 data pin. Address and Data fields are transferred at SDR using 2 data pins. Read data is sampled with CK. | 0x048 | — |
| 2S-2S-2S | Command, Address and Data fields are transferred at SDR using 2 data pins. Read data is sampled with CK. | 0x049 | — |
| 1S-4S-4S | Command field is transferred at SDR using 1 data pin. Address and Data fields are transferred at SDR using 4 data pins. Read data is sampled with CK. | 0x090 | — |
| 4S-4S-4S | Command, Address, and Data fields are transferred at SDR using 4 data pins. Read data is sampled with CK. | 0x092 | — |

Note: In XiP mode enable, XiP code is inserted in the Latency field. It is valid only for memory-mapping mode.

The bytes of Command and Address fields are transferred in the highest order to the lowest order sequence. The sequential bytes of the Data field are transferred in the lowest address to the highest address order. When using multiple pins, the least significant bit of each byte is placed on OM_SIO0 with each higher-order bit on the successively higher numbered OM_SIOn (n = 1 to 7) signals.

Figure 45.3 shows the timing diagram for 1S-1S-1S protocol mode. The OM_SIO0 signal is used for output data and the OM_SIO1 signal is used for input data.

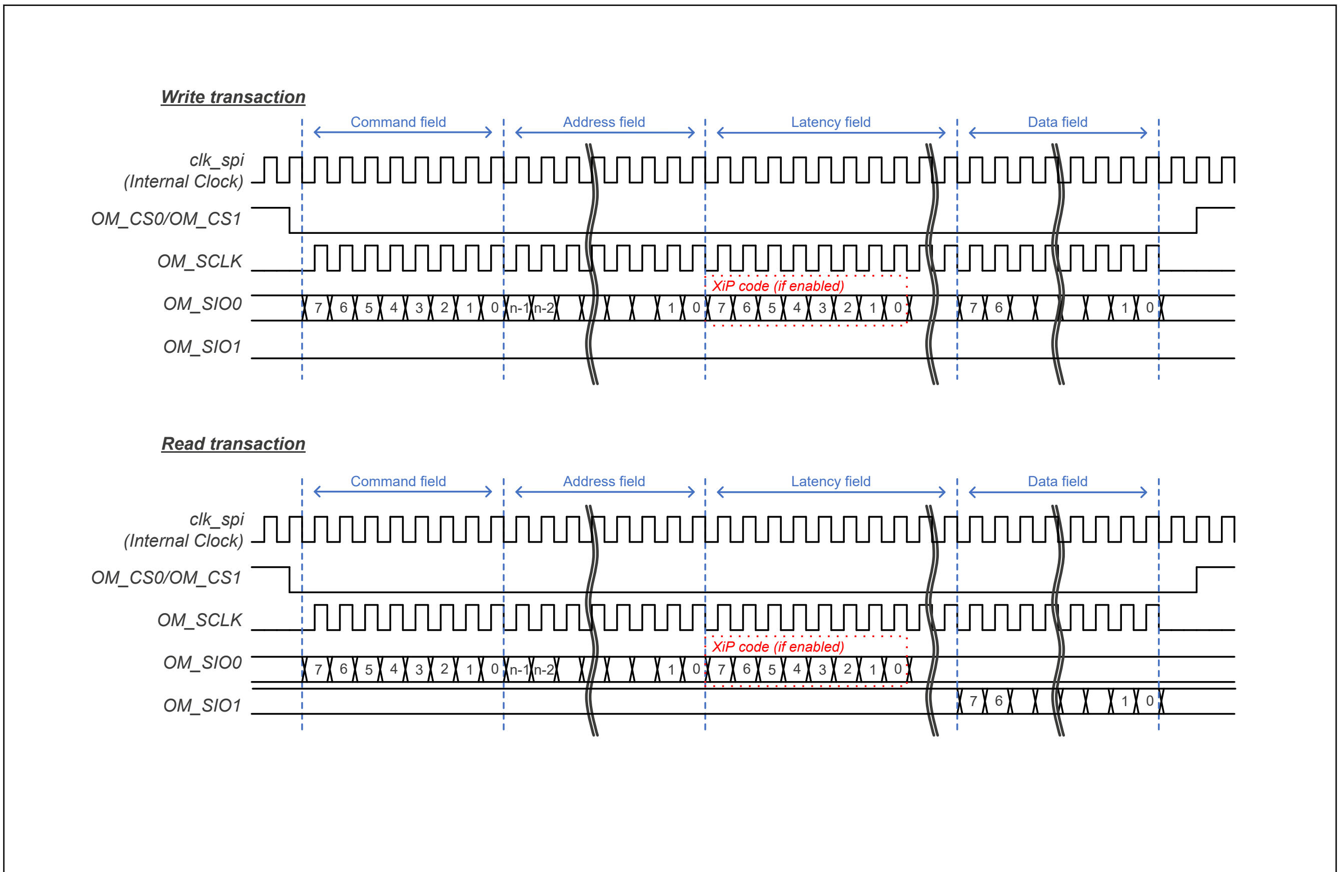


**Figure 45.3 1S-1S-1S timing diagram**

Figure 45.4 shows timing diagram for 1S-2S-2S protocol mode.

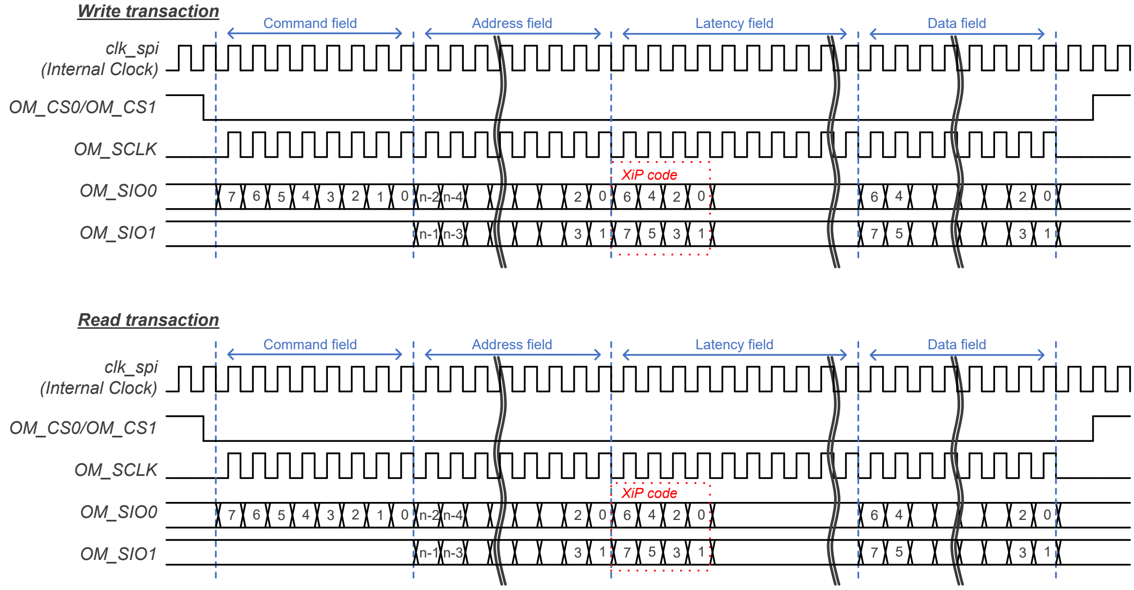


**Figure 45.4 1S-2S-2S timing diagram**

Figure 45.5 shows timing diagram for 4S-4D-4D protocol mode.

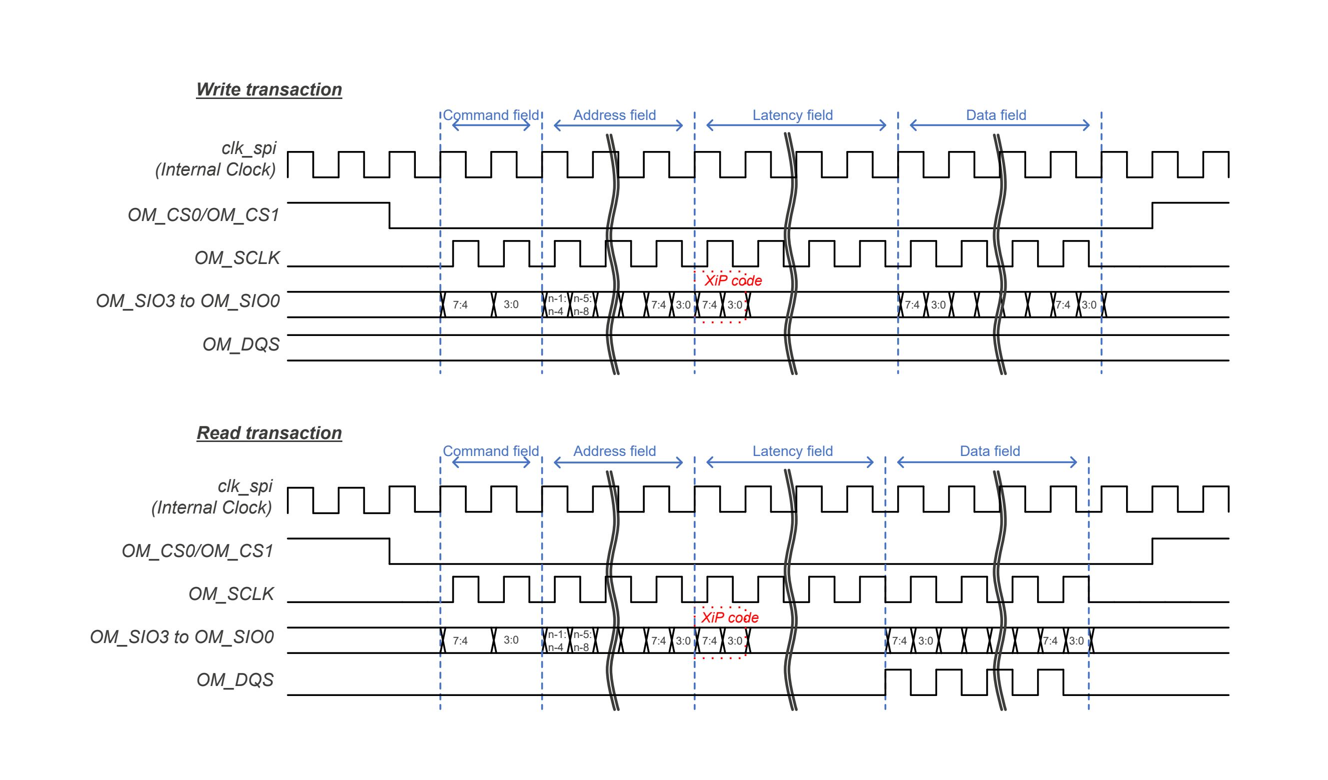


**Figure 45.5 4S-4D-4D timing diagram**

Figure 45.6 shows timing diagram for 8D-8D-8D profile 1.0 protocol mode. In 8D-8D-8D, the data is always transmitted with byte-pair. When each field is an odd byte, the last one byte is padded with invalid data.

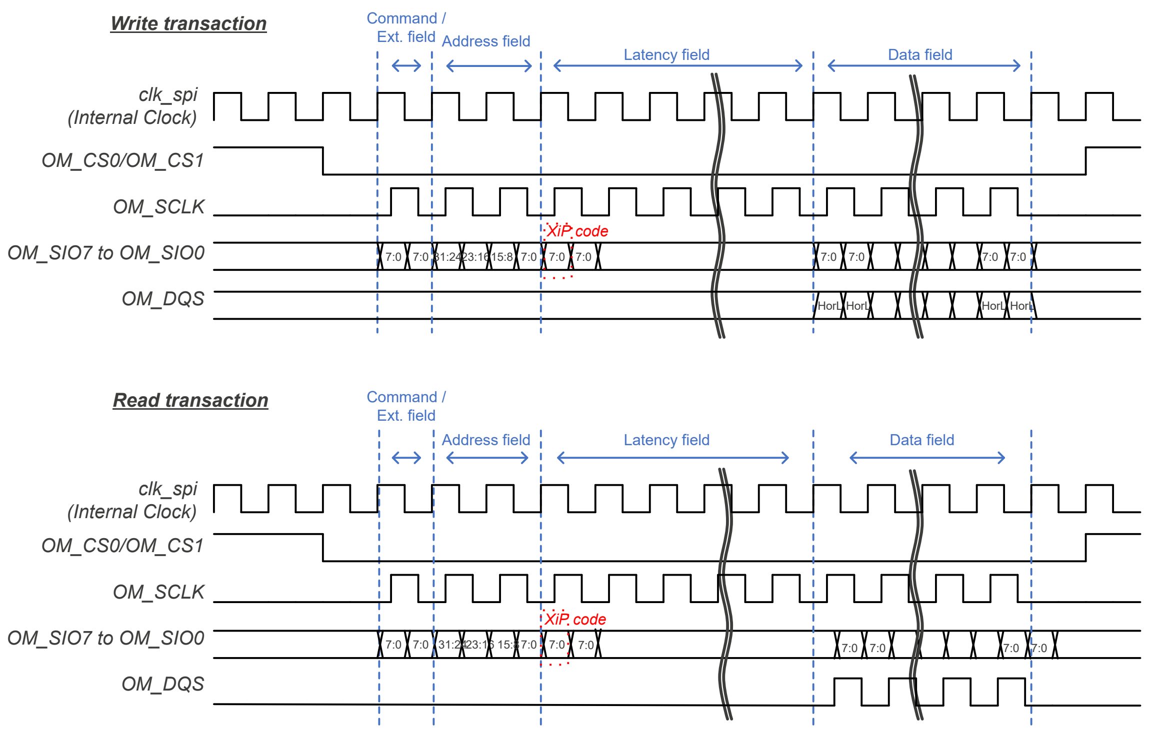


**Figure 45.6 8D-8D-8D profile 1.0 timing diagram**

Figure 45.7 shows timing diagram for 8D-8D-8D profile 2.0 protocol mode. In 8D-8D-8D, the data is always transmitted with byte-pair. When each field is an odd byte, the last one byte is padded with invalid data.

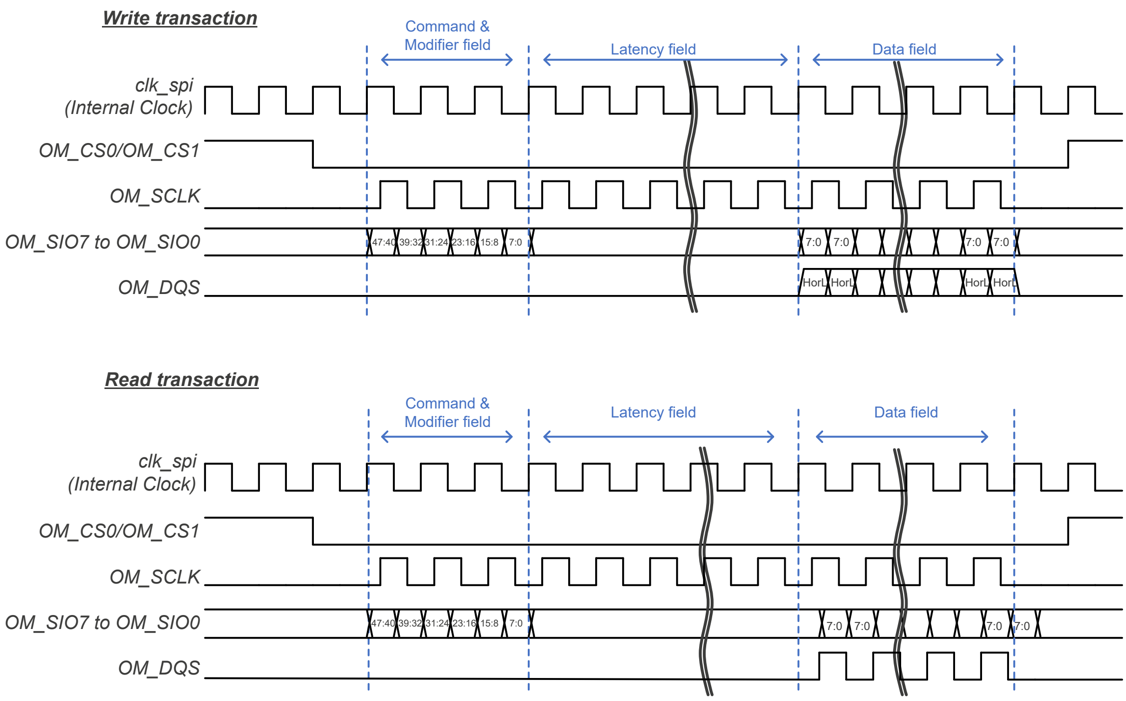


**Figure 45.7 8D-8D-8D profile 2.0 timing diagram**

## 45.3.1.2 xSPI Frame Interval

The interval between xSPI frames is configured with CS minimum idle term bits (LIOCFGCSn.CSMIN[3:0]). It depends on the specification of xSPI slave device.

## 45.3.1.3 OSPI Signals Timing Control

This xSPI Master supports both SDR and DDR. It is possible to sample input data with Data-Strobe (DS) signal at SDR. For various modes and easy implementation, this xSPI Master can adjust the timing to drive/sample xSPI interface signals statically. Table 45.4 shows the summary of xSPI interface signal timing control.

**Table 45.4 Summary of OSPI signals timing control (1 of 2)**

| Signal | Mode | Default operation | Timing control (n = 0, 1) |
|---|---|---|---|
| OM_CS0, OM_CS1 drive | — | Asserting 1 cycle before the rising edge of first OM_SCLK | 1 cycle extension for asserting with LIOCFGCSn.CSASTEX bit |
| | | Negating 1.5 cycle after the falling edge of last OM_SCLK | 1 cycle extension for negating with LIOCFGCSn.CSNEGEX bit |
| OM_SCLK drive | SDR without DS<br>SDR with DS | Reference point | — |
| | DDR with DS | Reference point | — |

**Table 45.4 Summary of OSPI signals timing control (2 of 2)**

| Signal | Mode | Default operation | Timing control (n = 0, 1) |
|---|---|---|---|
| OM_SIOn (n = 0 to 7) output drive | SDR without DS<br>SDR with DS | Falling edge of clk_spi (Internal Clock) | 0 or 0.5 cycle shift with LIOCFGCSn.SDRDRV bit |
| | DDR with DS | Both edges of clk_spi (internal clock) | — |
| OM_SIOn (n = 0 to 7) input sample | SDR without DS | Falling edge of OM_SCLK on expected data size | 0 to 7 cycle shift (1 cycle unit) with LIOCFGCSn.SDRSMPSFT[3:0] bits<br>0 or 0.5 cycle shift with LIOCFGCSn.SDRSMPMD bit |
| | SDR with DS | Falling edge of OM_DQS signal on expected data size | Sample at rising edge with LIOCFGCSn.SDRSMPMD bit<br>0 to 1 cycle phase shift with WRAPCFG.DSSFTCSn[4:0] bits<br>0 to 7 cycle extension with LIOCFGCSn.DDRSMPEX[3:0] bits |
| | DDR with DS | Both edges of OM_DQS signal on expected data size | 0 to 1 cycle phase shift with WRAPCFG.DSSFTCSn[4:0] bits<br>0 to 7 cycle extension with LIOCFGCSn.DDRSMPEX[3:0] bits |

Note: In DDR on xSPI protocol, DS should be aligned for the center of data. It means to shift the phase by 0.25 cycle (90 degrees). This xSPI master supports the adjustment of this phase depending on the usage conditions.

Figure 45.8 shows the default operation and timing control for SDR without DS.

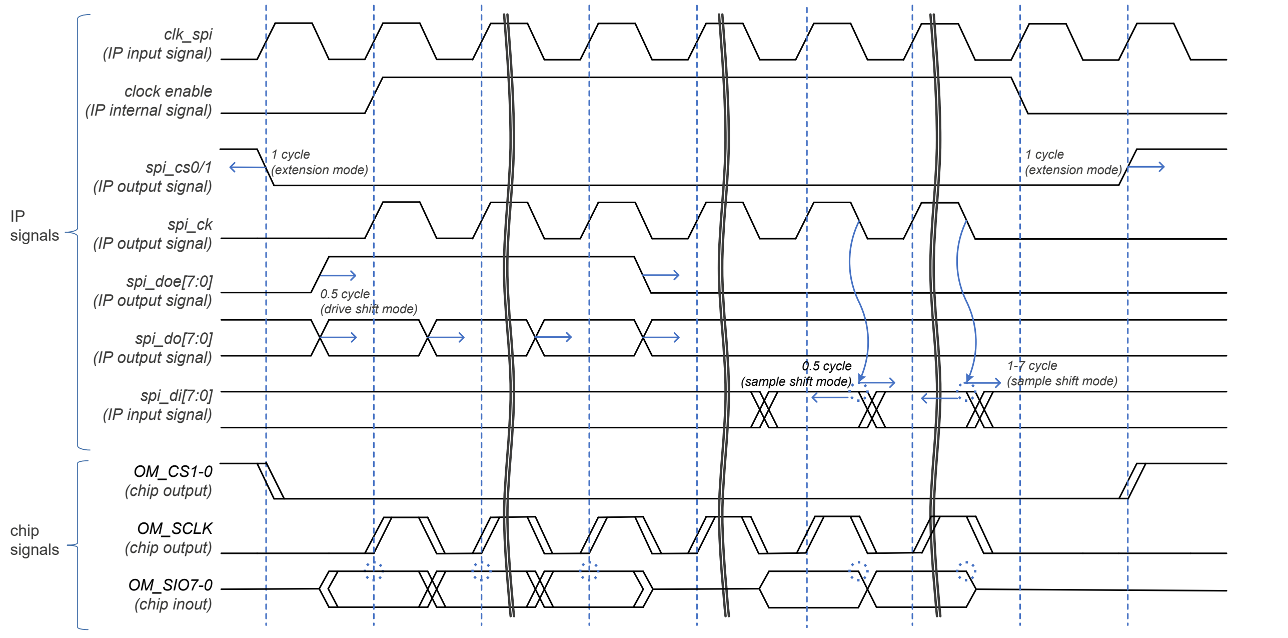


**Figure 45.8 Timing control for SDR without DS**

Figure 45.9 shows the default operation and timing control for SDR with DS.

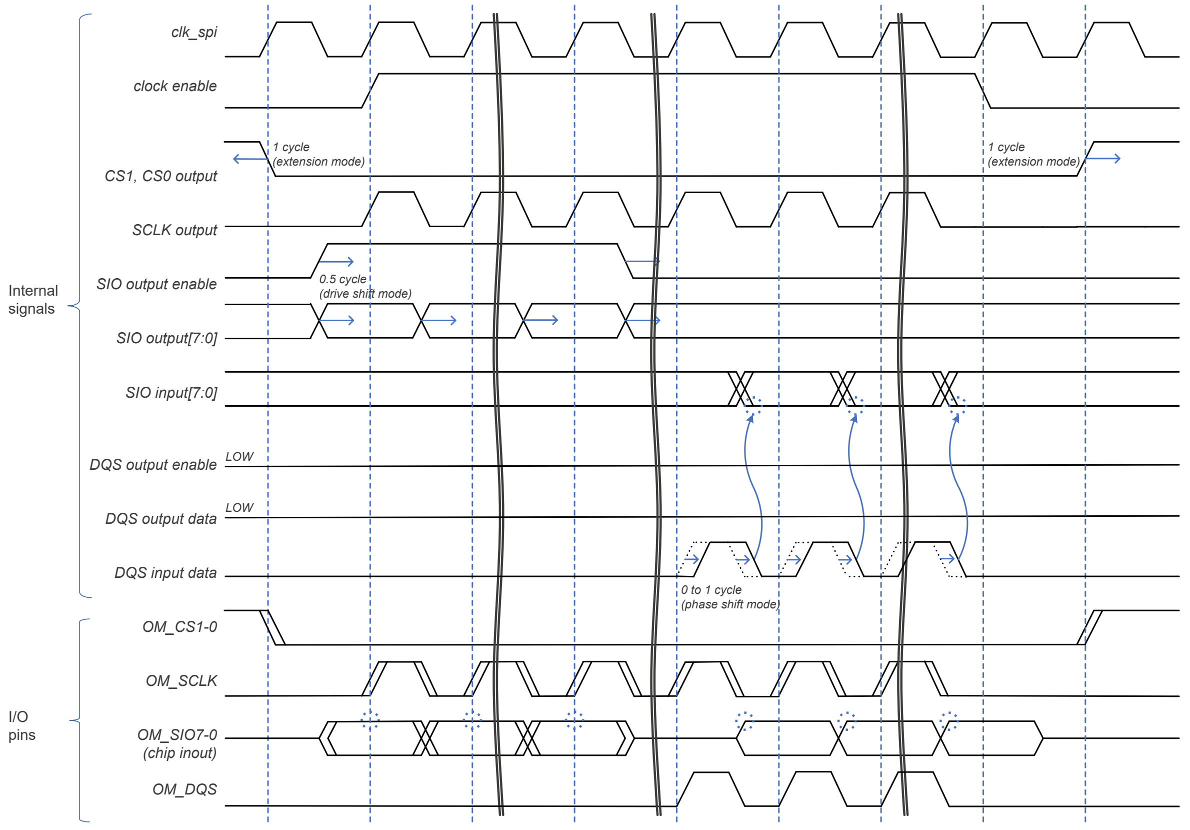


**Figure 45.9 Timing control for SDR with DS**

Figure 45.10 shows the default operation and timing control for DDR with DS.

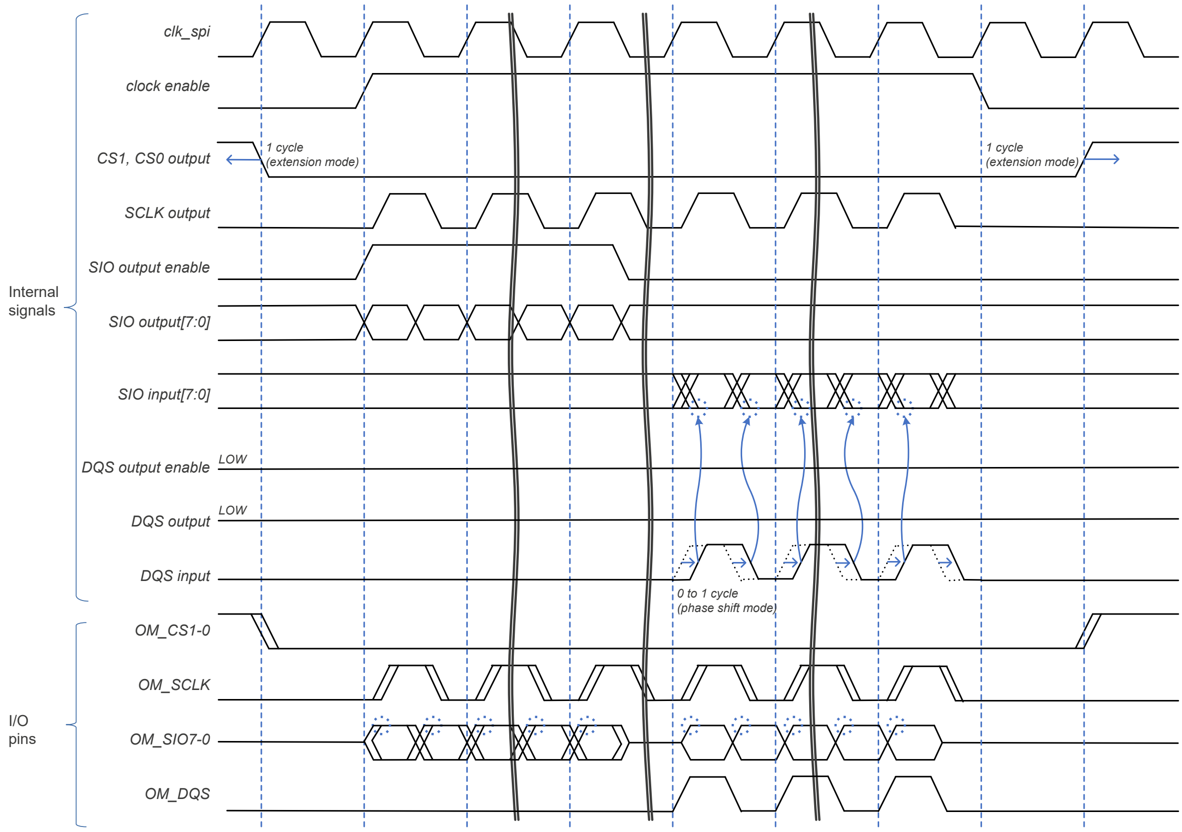


**Figure 45.10 Timing control for DDR with DS**

## 45.3.1.4 Automatic Calibration

The xSPI master supports the function to adjust DS shift value (WRAPCFG.DSSFTCSn (n = 0, 1)) automatically. When this function is enabled (CCCTL0CSn.CAEN = 1), this xSPI master transmits the calibration sequence periodically and adjusts the value of phase shift.

When all read compare is mismatched in read transaction during the calibration sequence, the calibration fail bit (INTS.CAFAILCSn = 1 (n = 0, 1)) is asserted and DS shift value is not updated. When at least one read compare is matched, the calibration success bit (INTS.CASUCCSn = 1 (n = 0, 1)) is asserted and DS shift value is updated. The result of each DS shift value can be monitored by the Calibration Status register (CASTTCSn). Figure 45.11 shows automatic calibration.

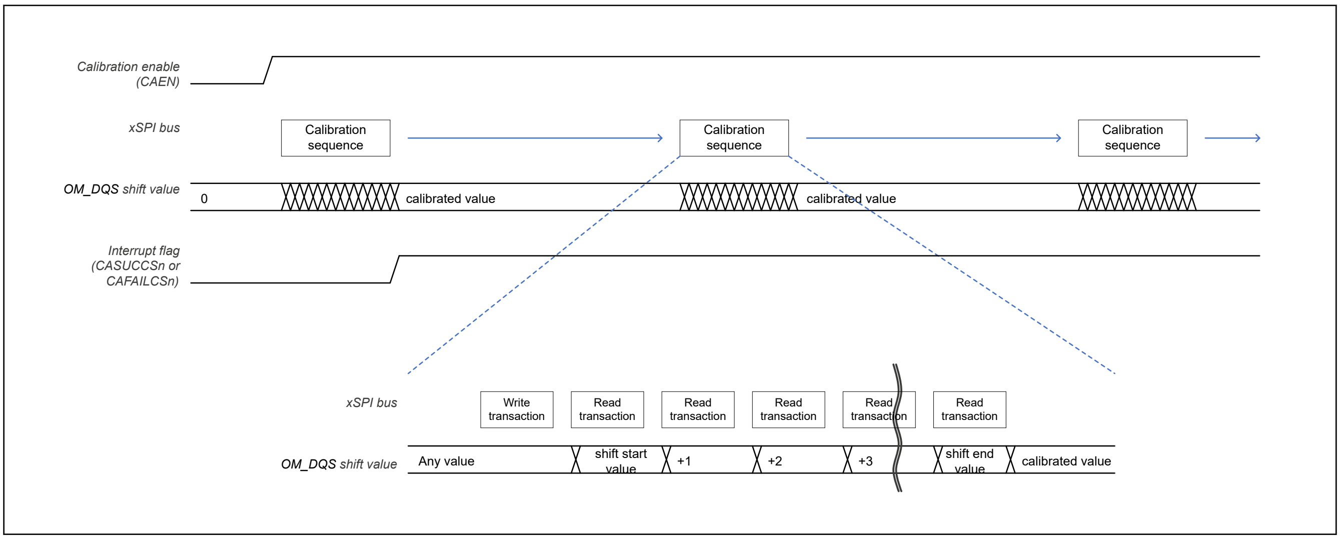


**Figure 45.11 Automatic calibration**

## 45.3.2 Manual-command

This section describes the manual-command mode. The manual-command has two functional modes:

- Direct mode
- Periodic mode.

### 45.3.2.1 Direct Mode

This mode sequentially can issue up to four xSPI transactions configured and requested by software. A series of transaction can be issued by a transaction request (CDCTL0.TRREQ = 1 with PERMD = 0). The number of transactions (CDCTL0.TRNUM[1:0]) can be configured up to 4. It can be used to change the mode or read the status of an xSPI slave device.

Table 45.5 shows the configured register bits for direct manual-command. Figure 45.25 shows the operating flow.

**Table 45.5 Manual-command configuration for direct mode**

| Transaction | Transaction type | Command | Command size | Address | Address size | Data (x = 0, 1) | Data size | Latency cycle |
|---|---|---|---|---|---|---|---|---|
| 1st Transaction | CDTBUF0. TRTYPE | CDTBUF0. CMD[15:0] | CDTBUF0. CMDSIZE [1:0] | CDABUF0. ADD[31:0] | CDTBUF0. ADDSIZE [2:0] | CDDxBUF0. DATA[31:0] | CDTBUF0. DATASIZE [3:0] | CDTBUF0. LATE[4:0] |
| 2nd Transaction | CDTBUF1. TRTYPE | CDTBUF1. CMD[15:0] | CDTBUF1. CMDSIZE [1:0] | CDABUF1. ADD[31:0] | CDTBUF1. ADDSIZE [2:0] | CDDxBUF1. DATA[31:0] | CDTBUF1. DATASIZE [3:0] | CDTBUF1. LATE[4:0] |
| 3rd Transaction | CDTBUF2. TRTYPE | CDTBUF2. CMD[15:0] | CDTBUF2. CMDSIZE [1:0] | CDABUF2. ADD[31:0] | CDTBUF2. ADDSIZE [2:0] | CDDxBUF2. DATA[31:0] | CDTBUF2. DATASIZE [3:0] | CDTBUF2. LATE[4:0] |
| 4th Transaction | CDTBUF3. TRTYPE | CDTBUF3. CMD[15:0] | CDTBUF3. CMDSIZE [1:0] | CDABUF3. ADD[31:0] | CDTBUF3. ADDSIZE [2:0] | CDDxBUF3. DATA[31:0] | CDTBUF3. DATASIZE [3:0] | CDTBUF3. LATE[4:0] |

### 45.3.2.2 Periodic Mode

This mode periodically issues an xSPI read transaction configured and requested by software, and can compare the read value up to 4 bytes with the expected value. The transaction is issued by a transaction request (CDCTL0.TRREQ = 1 with PERMD = 1). It can be used to alternate the status polling operation of an xSPI slave device.

The periodic term is configured in the periodic transaction interval bits (CDCTL0.PERITV[4:0]). The expected value is configured in the periodic transaction expected and masked value bits (CDCTL1.PEREXP[31:0] and CDCTL2.PERMSK[31:0]). Table 45.6 shows the configured register bits for periodic manual-command. Figure 45.26 shows the operating flow.

**Table 45.6 Manual-command configuration for periodic mode**

| Transaction | Transaction type | Command | Command size | Address | Address size | Data (x = 0, 1) | Data size | Latency cycle |
|---|---|---|---|---|---|---|---|---|
| Read Transaction | CDTBUF0. TRTYPE | CDTBUF0. CMD[15:0] | CDTBUF0. CMDSIZE [1:0] | CDABUF0. ADD[31:0] | CDTBUF0. ADDSIZE [2:0] | CDDxBUF0. DATA[31:0] | CDTBUF0. DATASIZE [3:0] | CDTBUF0. LATE[4:0] |

## 45.3.3 Memory-mapping

This section describes the memory-mapping mode. This mode automatically converts system bus access for a preconfigured memory area into an xSPI transaction.

### 45.3.3.1 Configuration

In this operation, the payload of the Address and Data fields are delivered from system bus signals. The information of the Command field and size are delivered from the configured register bits. Table 45.7 shows the register bits configured for memory-mapping.

**Table 45.7 Memory-mapping configuration for memory area access (n = 0, 1)**

| System bus transaction | Format change mode | Command | Command size | Address | Address size | Data | Data size | Latency cycle |
|---|---|---|---|---|---|---|---|---|
| Write for slave n memory area | Normal | CMCFG2CSn. WRCMD[15:8] | 1 byte | SAWADDR[x:0] | CMCFG0CSn. ADDSIZE [1:0] | SWDATA | Up to SAWLEN and SAWSIZE | CMCFG2CSn. WRLATE[4:0] |
| | 8D-8D-8D profile 1.0 | CMCFG2CSn. WRCMD[15:0] | 2 bytes | | | | | |
| | 8D-8D-8D profile 2.0 Command Modifier | CMCFG2CSn. WRCMD[15:8] | 1 byte | {SAWADDR[27:4], 0000000000000b, SAWADDR[3:1]} | 5 bytes | | | |
| | 8D-8D-8D profile 2.0 Extended Command Modifier | CMCFG2CSn. WRCMD[15:13] | 3 bits | {0b, SAWADDR [31:4], 0000000000000b, SAWADDR[3:1] | 45 bits | | | |
| Read for slave n memory area | Normal | CMCFG1CSn. RDCMD[15:8] | 1 byte | SARADDR[x:0] | CMCFG0CSn. ADDSIZE [1:0] | SRDATA | Up to SARLEN and SARSIZE | CMCFG1CSn. RDLATE[4:0] |
| | 8D-8D-8D profile 1.0 | CMCFG1CSn. RDCMD[15:0] | 2 bytes | | | | | |
| | 8D-8D-8D profile 2.0 Command Modifier | CMCFG1CSn. RDCMD[15:8] | 1 byte | {SARADDR[27:4], 0000000000000b, SARADDR[3:1]} | 5 bytes | | | |
| | 8D-8D-8D profile 2.0 Extended Command Modifier | CMCFG1CSn. RDCMD[15:13] | 3 bits | {0b, SARADDR [31:4], 0000000000000b, SARADDR[3:1]} | 45 bits | | | |

Note: The MSByte of Address can be replaced with Address Replace Enable and Code bits (CMCFG0CSn.ADDRPEN[7:0]/ADDRPCD[7:0]).

### 45.3.3.2 Write Access Operation

At accepting write access for memory area from system bus, this xSPI Master stores all payload data in an internal bridge buffer and then issues a write transaction into an xSPI slave. Figure 45.12 shows the operation summary.

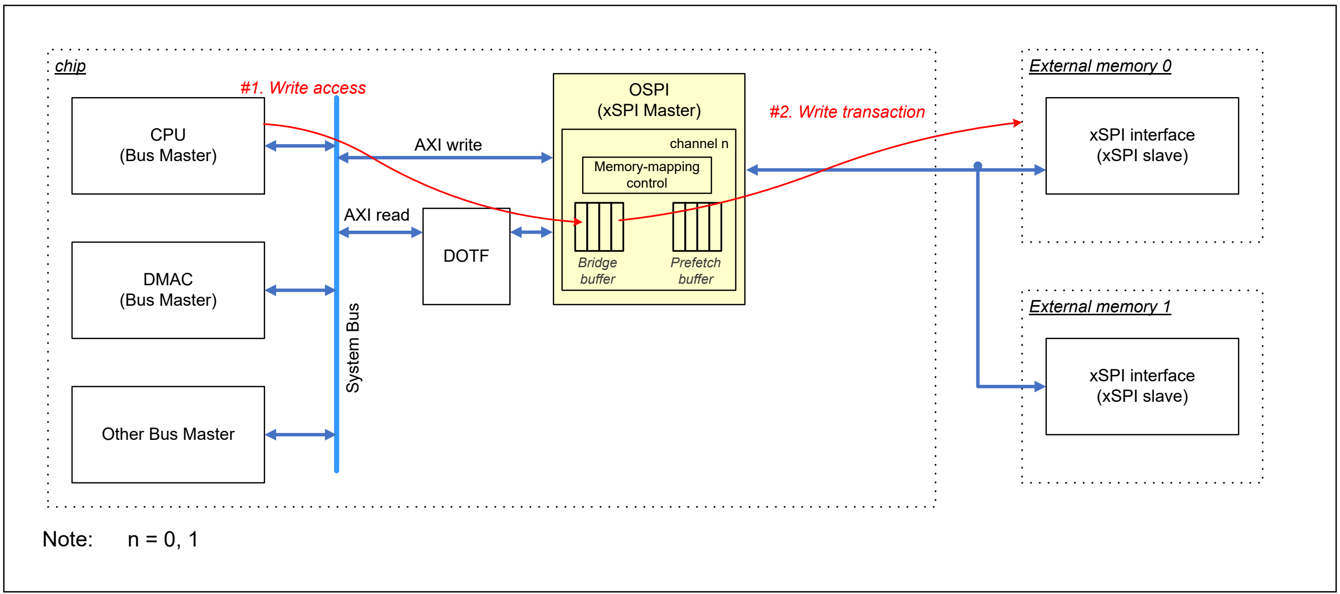


**Figure 45.12 Write access for memory area**

The operation of xSPI bus changes depending on the burst type of the system bus. When the burst type is single type or incremental type, one system bus transaction triggers one xSPI frame. When the burst type is wrap type and the CMCFG0CSn.WPBSTMD is 0, one system bus transaction triggers two xSPI frames. Figure 45.13 shows the relationship between AXI and xSPI frames.

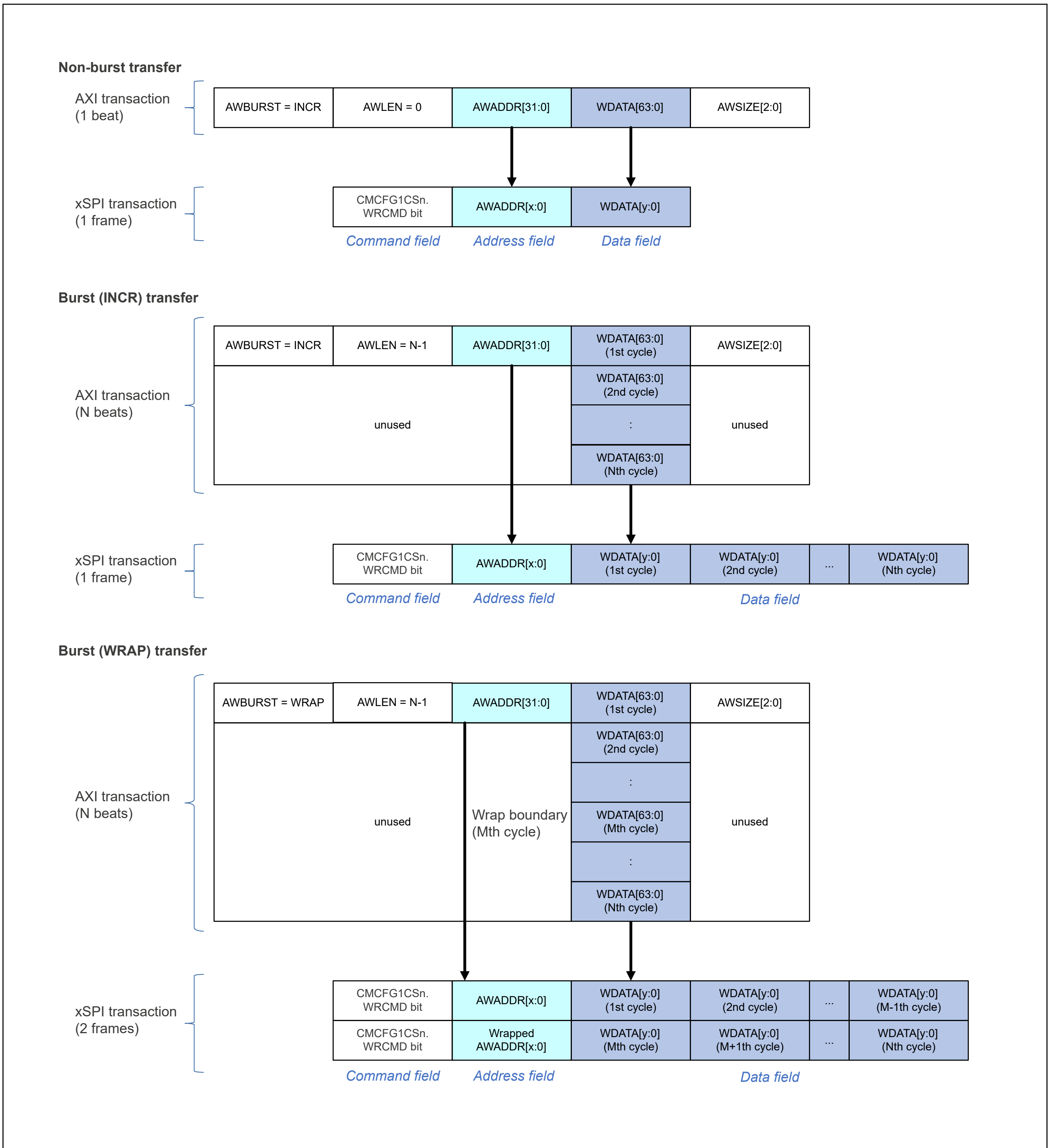


**Figure 45.13 xSPI frame format in write access (Normal format)**

### 45.3.3.3 Combination Function

At the system bus write access for memory area, this xSPI master has the function to combine the write data for high throughput on the xSPI bus. When this function is enabled (BMCFGCHn.MWRCOMB = 1), this xSPI master transmits a xSPI frame with the selected size while the sequential address is incremental*1. When one of the below conditions is detected, even though not reaching to the target size (BMCFGCHn.MWRSIZE[7:0]), this xSPI master transmits the pending data into the xSPI bus.

- Nonincremental address is detected
- Different burst type is detected

- Read transaction is detected
- Access for different slave is detected
- Memory Write Data Push bit (BMCTL1.MWRPUSHCHn (n = 0, 1)) is set.

This function is useful for any slave device to request a chunk of data at a time. In such case, the system bus master continues to provide the fixed data size with an incremental address. For example, there is a device to request to write in page unit.

Figure 45.14 shows the operation when enabling the combination function.

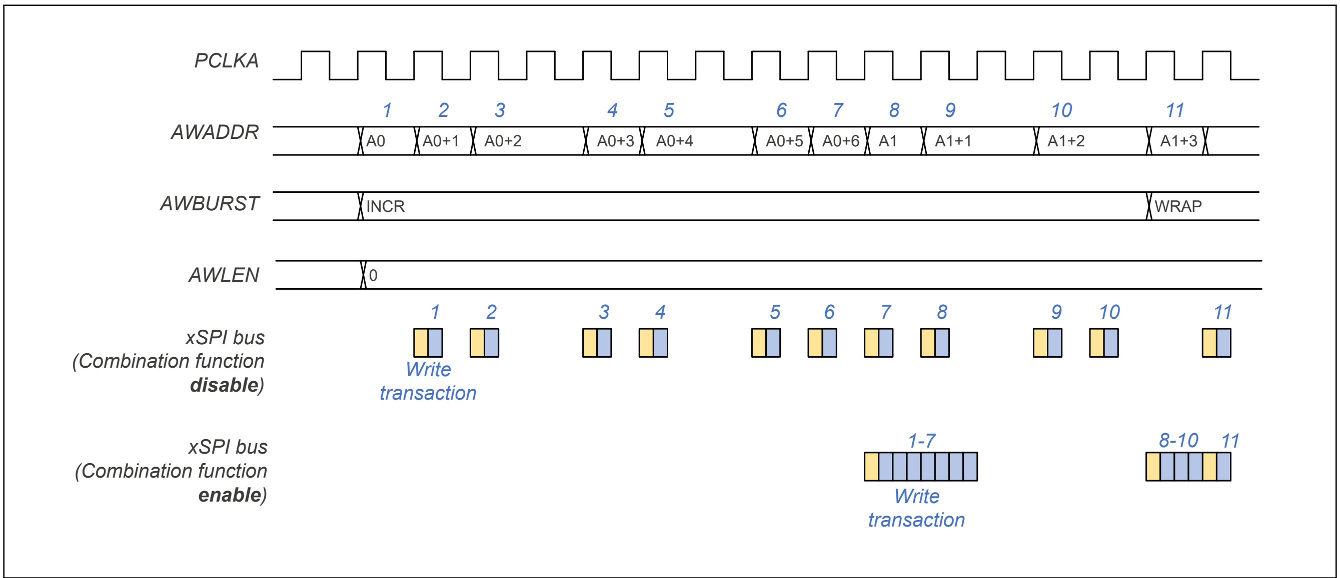


**Figure 45.14 Combination function**

Note 1. The access which complies with all the following conditions is considered as incremental address.

- Transaction type is INCR
- The start address of the access is continuous to the previous last write address.
  - The start position of the access for WSTRB is treated as the start address
  - The last WSTRB of the previous access is treated as the last write address.

Figure 45.15 shows a data combined example with AXI access.

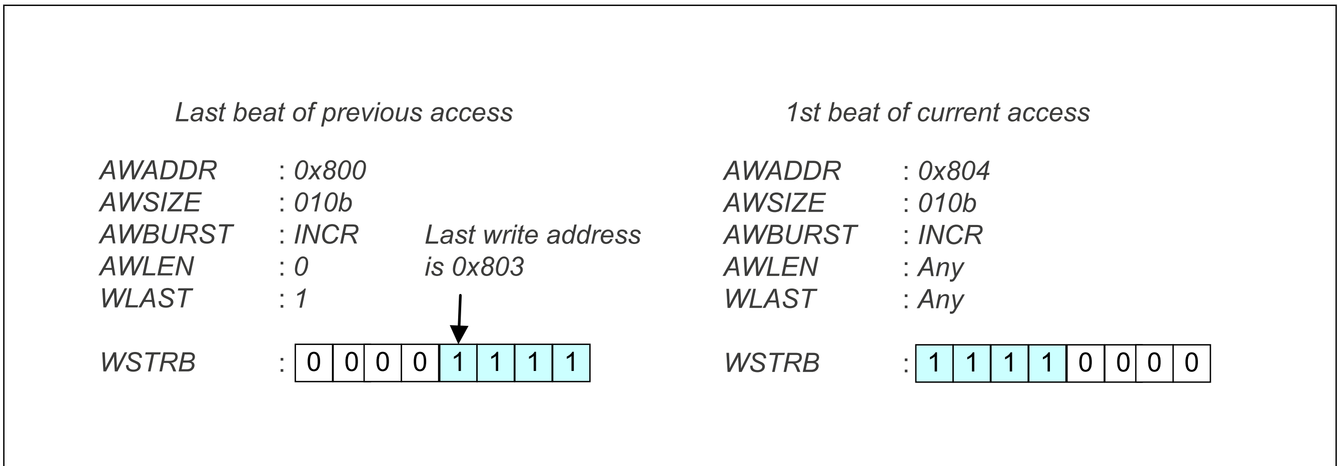


**Figure 45.15 Data combined example with AXI access**

### 45.3.3.4 Read Access Operation

At the read access for memory area, soon after detected the read access, this xSPI Master issues a read transaction into an xSPI slave. Figure 45.16 shows the operation summary.

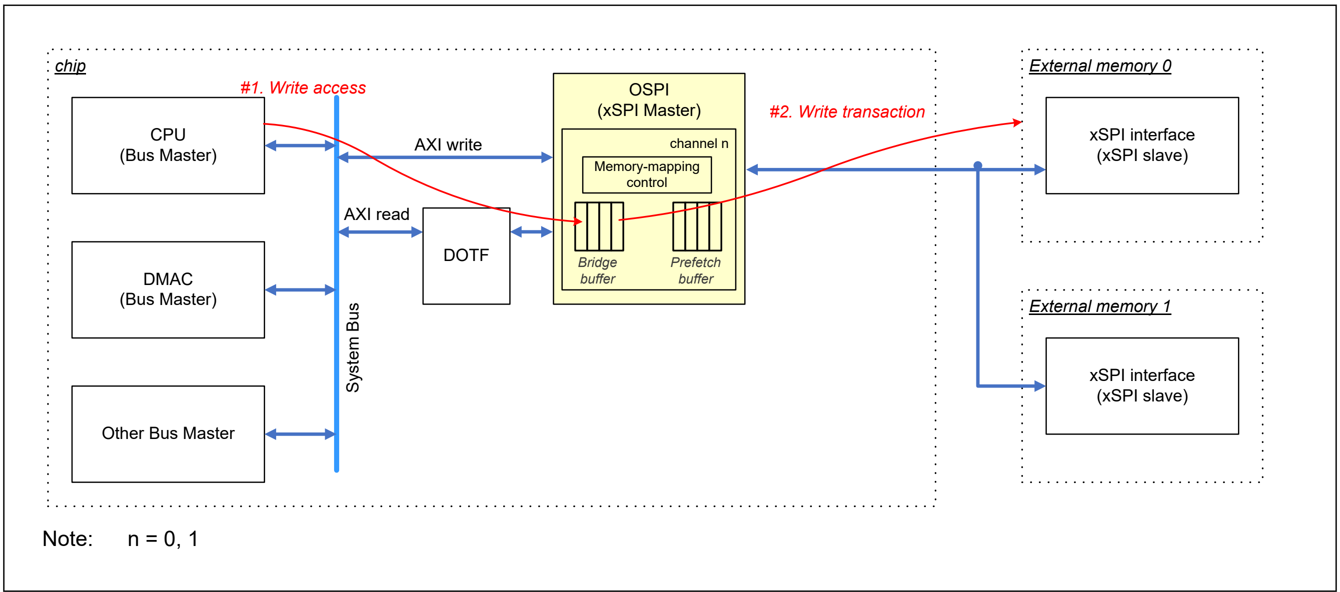


**Figure 45.16 Read access for memory area**

The operation of xSPI bus changes depending on the burst type. When the type is single or increment type, one read transaction of a system bus triggers one xSPI frame. When the type is wrap type and the CMCFG0CSn.WPBSTMD is 0, one read transaction of a system bus triggers two xSPI frames. Figure 45.17 shows the relationship between AXI and xSPI frames.

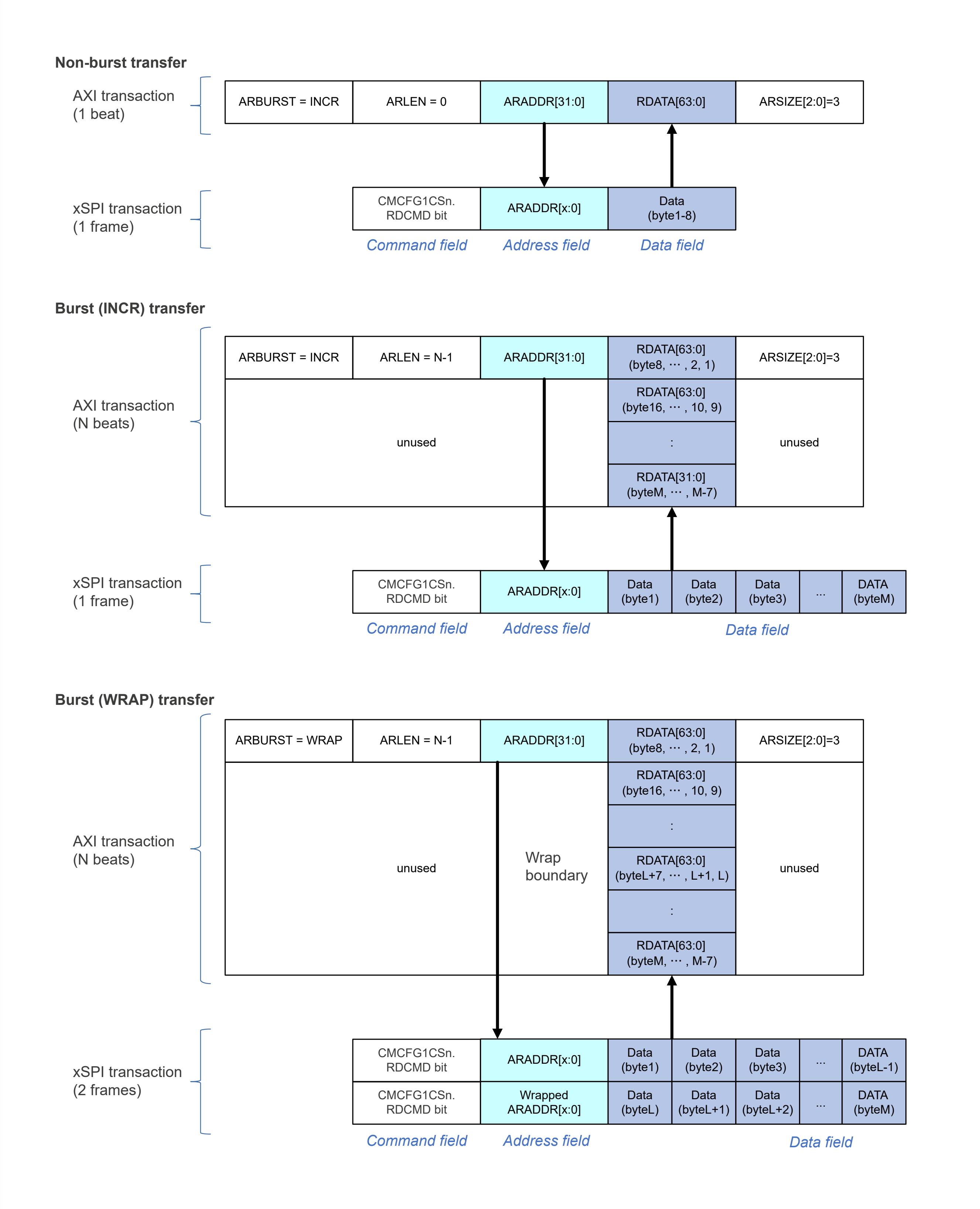


**Figure 45.17 xSPI frame format in read access (Normal format)**

### 45.3.3.5 Prefetch Function

At the read access for memory area from the system bus, this xSPI Master has the function to prefetch read data for reducing the latency. When enabled this function (BMCFGCHn.PREEN = 1), this xSPI Master continues to read the incremental address and store the read data from an xSPI slave in internal prefetch buffer. This xSPI master then searches in the prefetch buffer for the following read access from the system bus. If the target read data in the prefetch buffer is found, this xSPI Master returns the data from the prefetch buffer. If not found, this xSPI Master clears the prefetch buffer and issues a new read transaction into an xSPI slave. This function is effective in applications such as when the consecutive read addresses are close. But it is not effective in applications such as when the consecutive read addresses are not incremental because xSPI read frame for prefetch uses xSPI bus. Figure 45.18 shows the operation summary.

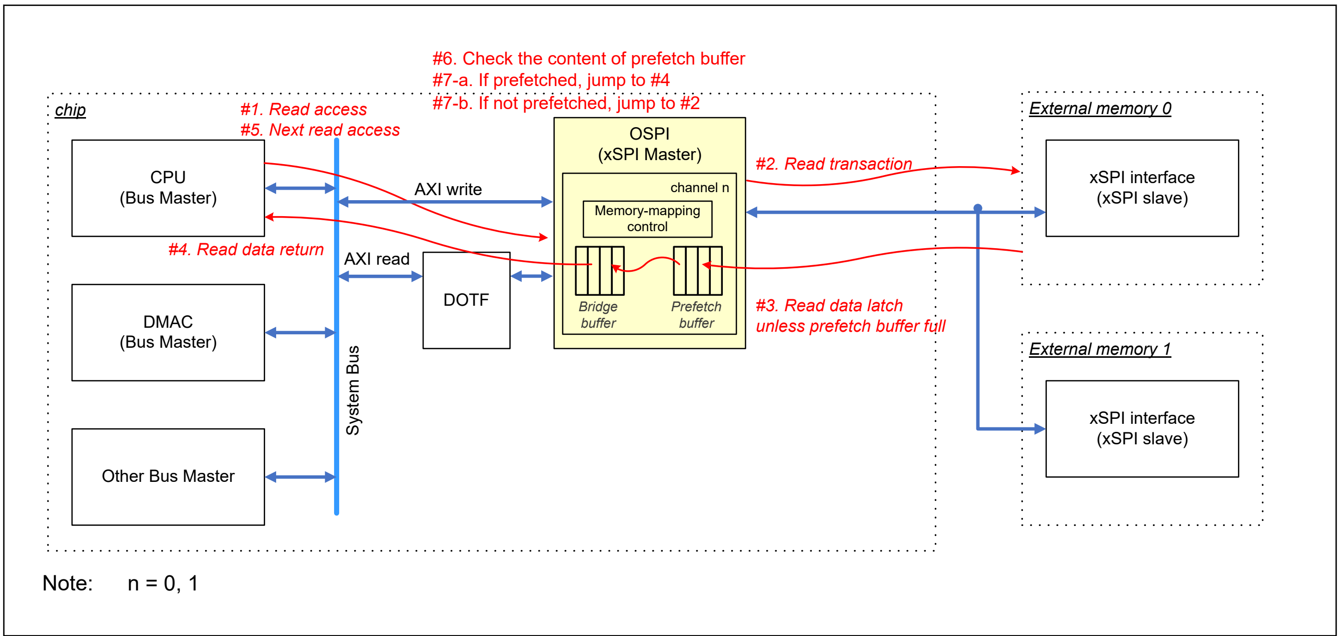


**Figure 45.18 Read access for memory area with prefetch enabled**

Note: When enabled this prefetch function, the bus master can read from the internal prefetch buffer but not from a slave device. When accessed to the same address from multiple bus masters, this xSPI Master does not guarantee to read the latest data. If the bus master wants to read the latest data from a slave device, it should read after clearing the prefetch buffer (BMCTL1.PBUFCLRCHn (n = 0, 1)).

Note: Prefetch buffer is implemented as FIFO-based, and when a read access is issued, the data before the access address is discarded from the buffer. When the next access is issued to the region which is discarded at the previous access, this module issues the xSPI read access again to fill the prefetch buffer.

Note: This IP has a 1-line buffer which keeps the last 8-byte read data from the prefetch buffer. When a read access is issued to the same address and to the data in the 1-line buffer, this IP returns read data from the 1-line buffer.

### 45.3.3.6 XiP Mode

Some slave devices have a mode (XiP mode) in which the command phase is not required for lower latency. While in this mode, the xSPI master skips sending the command and the slave device implicitly performs the command that was executed in the previous transaction. When enabled XiP mode bit (CMCTLCHn.XIPEN = 1), this xSPI master inserts XiP enter code (CMCTLCHn.XIPENCODE[7:0]) in the Latency field. When disabled XiP mode bit (CMCTLCHn.XIPEN = 0), this xSPI master inserts XiP exit code (CMCTLCHn.XIPEXCODE[7:0]) in the Latency field. This function is available only for memory-mapping mode.

When the xSPI master transmits the XiP disable pattern, this master clears the XiP mode bit and disables XiP mode configured for both channels. It is not possible to disable XiP mode for only one channel by transmitting a XiP disable pattern.

Note: When enough latency cycle does not exist for XiP code, this xSPI Master cannot insert XiP code.

Note: XiP mode can be used only for unidirectional access to a slave. The write transaction and read transaction should be separated.

Note: The XiP exit code is inserted once when disabled. For more details, see Figure 45.30.

### 45.3.4 Pattern Control

This xSPI Master has the function to transmit three types of pattern which are not xSPI frame format. The pattern is triggered by the setting trigger bit (LPCTL0-1.PATREQ).

#### 45.3.4.1 XiP Disable Pattern

XiP Disable pattern transmits any pattern with the configured length and value (LPCTL0.XD1LEN[4:0]/XD1VAL/XD2LEN[4:0]/XD2VAL). XiP Disable pattern uses OM_SCLK, OM_SIO7-0 signals. The number of output pins can be configured by the XiP Disable pattern pin bits (LPCTL0.XDPIN[1:0]). It may be used to disable XiP mode for legacy SPI. Figure 45.19 shows the timing diagram.

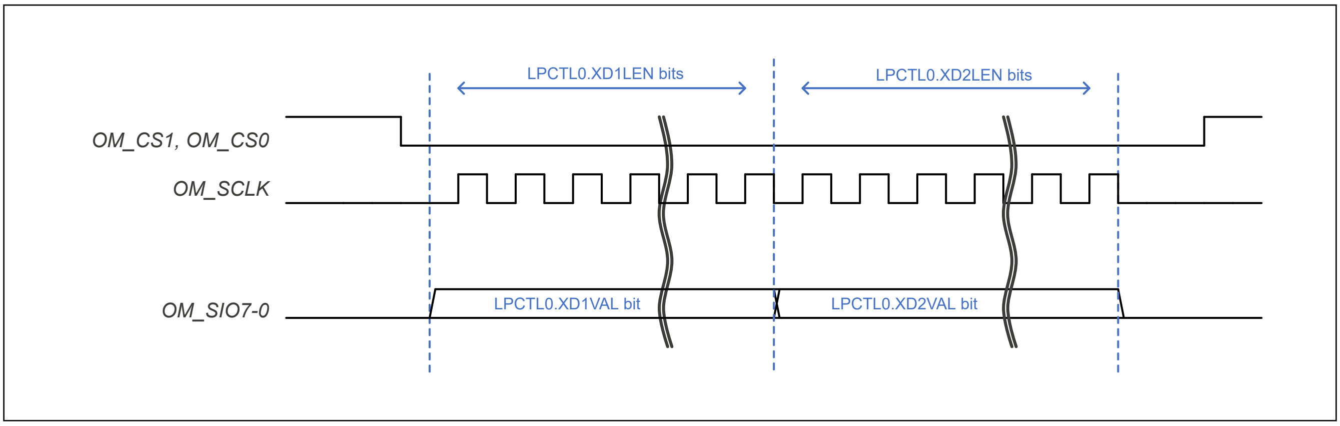


**Figure 45.19 XiP Disable pattern**

#### 45.3.4.2 Reset Pattern

Reset pattern transmits the pattern specified in the Serial Flash Reset Signaling Protocol. Figure 45.20 shows the timing diagram.

CS Low/High width is configured with the Reset Pattern Length bits (LPCTL1.RSTWID[2:0]). The xSPI slave samples the data input at the rising edge of CS. Setup time for data output is configured with Reset pattern data output setup time bits (LPCTL1.RSTSU[2:0]). The setup time should always be less than Reset pattern width.

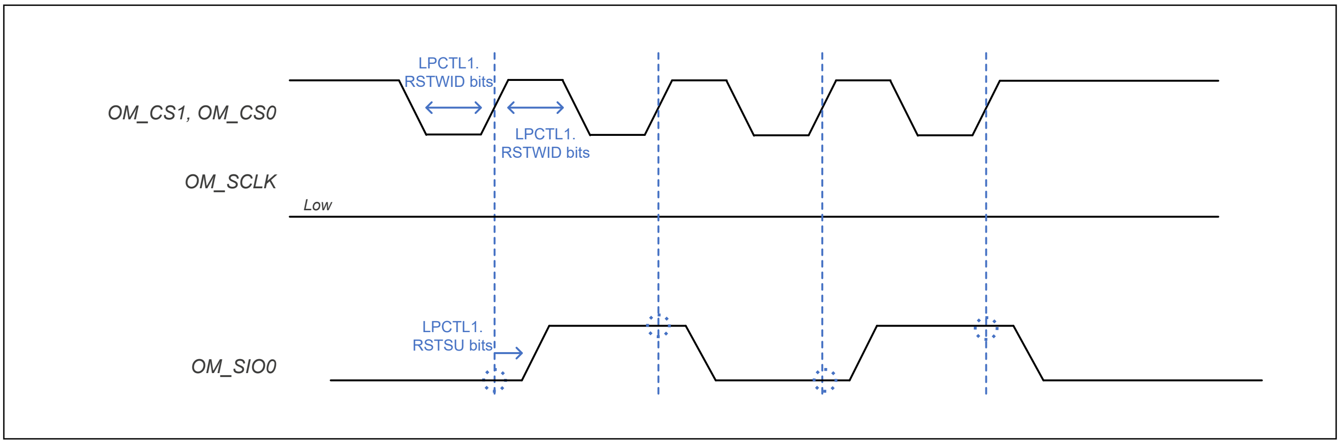


**Figure 45.20 Reset pattern**

Note: In the protocol, CS Low/High width is defined as minimum 500 ns and setup time is defined as minimum 6 ns.

#### 45.3.4.3 CS-only Pattern

CS-only pattern activates the CS port with the configured length bits (LPCTL1.RSTWID[2:0]). It may be used to resume from the Deep Power Down state. Figure 45.21 shows the timing diagram.

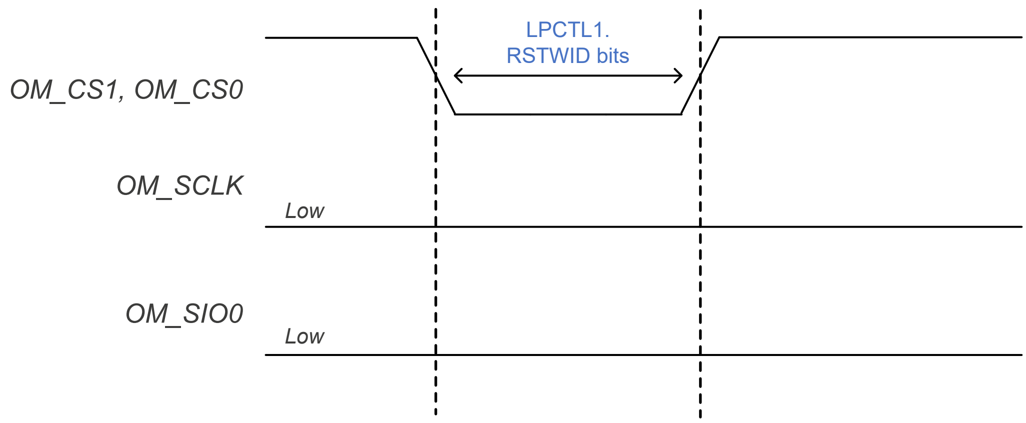


**Figure 45.21 CS-only pattern**

## 45.3.5 Integrity Checking

This xSPI Master can detect some errors. Table 45.8 shows the error list and the detail behavior.

**Table 45.8 Error list (n = 0, 1)**

| Error type | Event | Flag bit | Note (action) |
|---|---|---|---|
| Calibration failed | When the read data did not match the expected value during automatic calibration. | INTS.CAFAILCSn | It results in writing unexpected data to xSPI slave. |
| System bus error | When an error response occurred on the AXI slave interface for memory-mapping. | INTS.BUSERRCHn | This xSPI master should be reset for fatal error. |
| ECC error detection | When detected the falling edge on OM_ECSINT1 port.<br>It can be useful only for xSPI slave with ECC detection function. | INTS.ECSCS1 | Only notify the error event of xSPI slave. |
| DS timeout | When DS does not toggle in read transaction with using DS. | INTS.DSTOCSn | If this error occurs during calibration, clear the interrupt status bit.<br>Other than above, both xSPI master and xSPI slave should be reset for fatal error. |
| Periodic transaction timeout | When the read value does not match with the expected value in periodic manual-command mode. | INTS.PERTO | Depending on the status of the function. |

## 45.3.6 Interrupts

This xSPI Master has an interrupt port. It can monitor with the Interrupt Status Register (INTS). During the initialization phase, it can be programmable with the Interrupt Enable register (INTE). Table 45.9 shows OSPI interrupt sources, and Table 45.10 shows the related register bit.

Note: The interrupt pulse port signal is not asserted when the corresponding bit of INTE is set after the interrupt event is detected.

**Table 45.9 OSPI interrupt sources**

| Name | Interrupt sources | DMAC activation |
|---|---|---|
| OSPI0_ERR | Error | Not possible |
| OSPI0_CMP | Complete | Not possible |

**Table 45.10 Interrupt register bit**

| Flag bit | Enable bit | Clear bit | Interrupt sources |
|---|---|---|---|
| CASUCCS1 | CASUCCS1E | CASUCCS1C | OSPI0_CMP |
| CASUCCS0 | CASUCCS0E | CASUCCS0C | |
| CAFAILCS1 | CAFAILCS1E | CAFAILCS1C | OSPI0_ERR |
| CAFAILCS0 | CAFAILCS0E | CAFAILCS0C | |
| BUSERRCH1 | BUSERRCH1E | BUSERRCH1C | OSPI0_ERR |
| BUSERRCH0 | BUSERRCH0E | BUSERRCH0C | |
| INTCS1 | INTCS1E | INTCS1C | OSPI0_ERR |
| ECSCS1 | ECSCS1E | ECSCS1C | OSPI0_ERR |
| DSTOCS1 | DSTOCS1E | DSTOCS1C | OSPI0_ERR |
| DSTOCS0 | DSTOCS0E | DSTOCS0C | |
| PERTO | PERTOE | PERTOC | OSPI0_ERR |
| PATCMP | PATCMPE | PATCMPC | OSPI0_CMP |
| CMDCMP | CMDCMPE | CMDCMPC | OSPI0_CMP |

## 45.3.7 Flows of Operations

### 45.3.7.1 Flow of Configuration

Figure 45.22 shows flow of configuration.

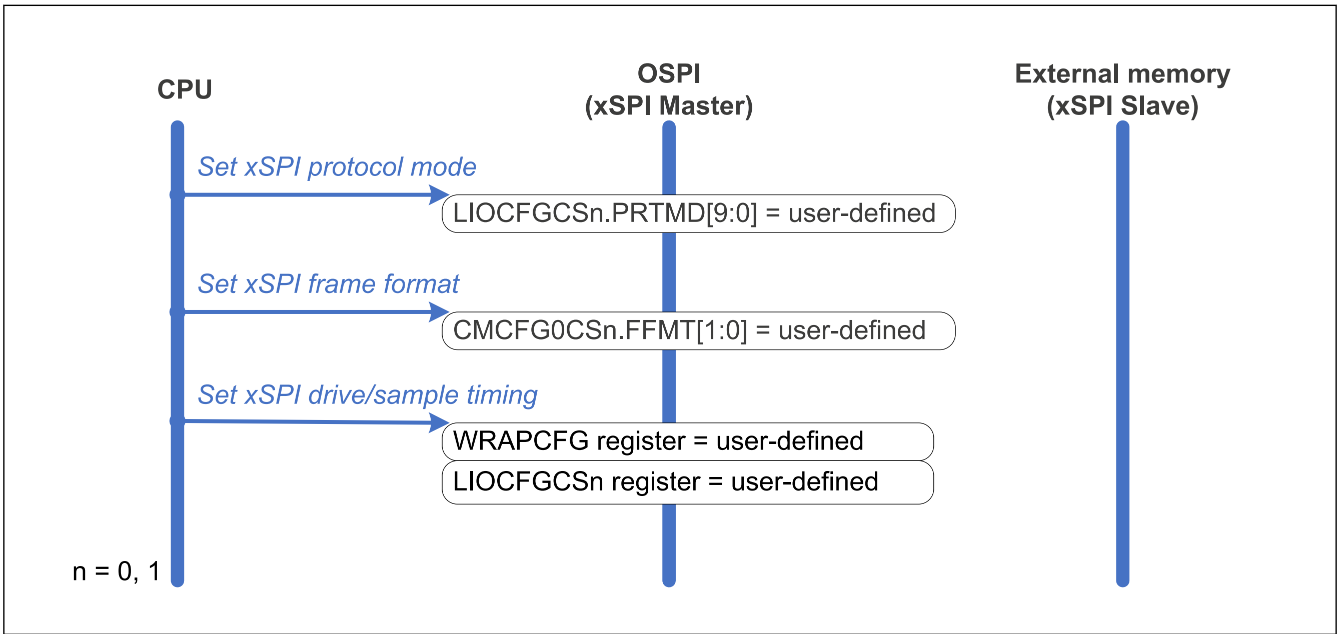


**Figure 45.22 Flow of configuration**

### 45.3.7.2 Flow of Communication Stop

Figure 45.23 shows the flow of a communication stop.

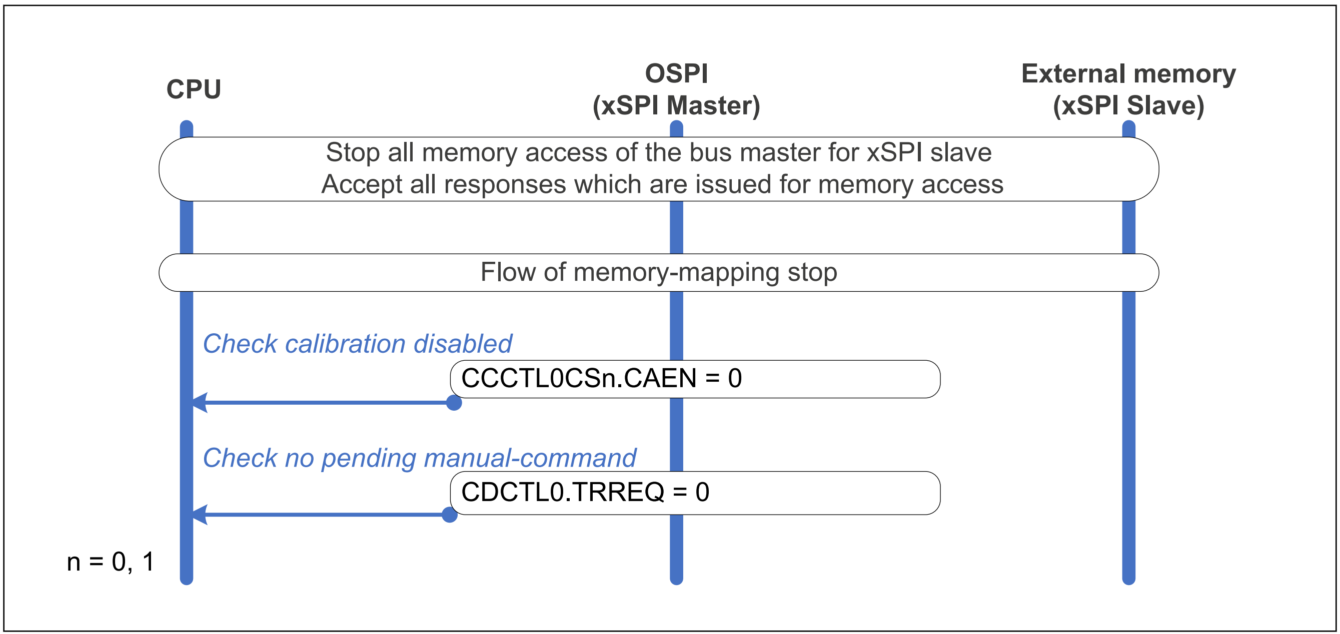


**Figure 45.23 Flow of communication stop**

Note: When reconfiguring any configuration register, stop all communication with the xSPI slave to avoid race condition between register setting and memory access. It means the automatic calibration is disabled, and there is no pending manual-command and memory-mapping access.

### 45.3.7.3 Flow of Automatic Calibration

Figure 45.24 shows the flow of an automatic calibration.

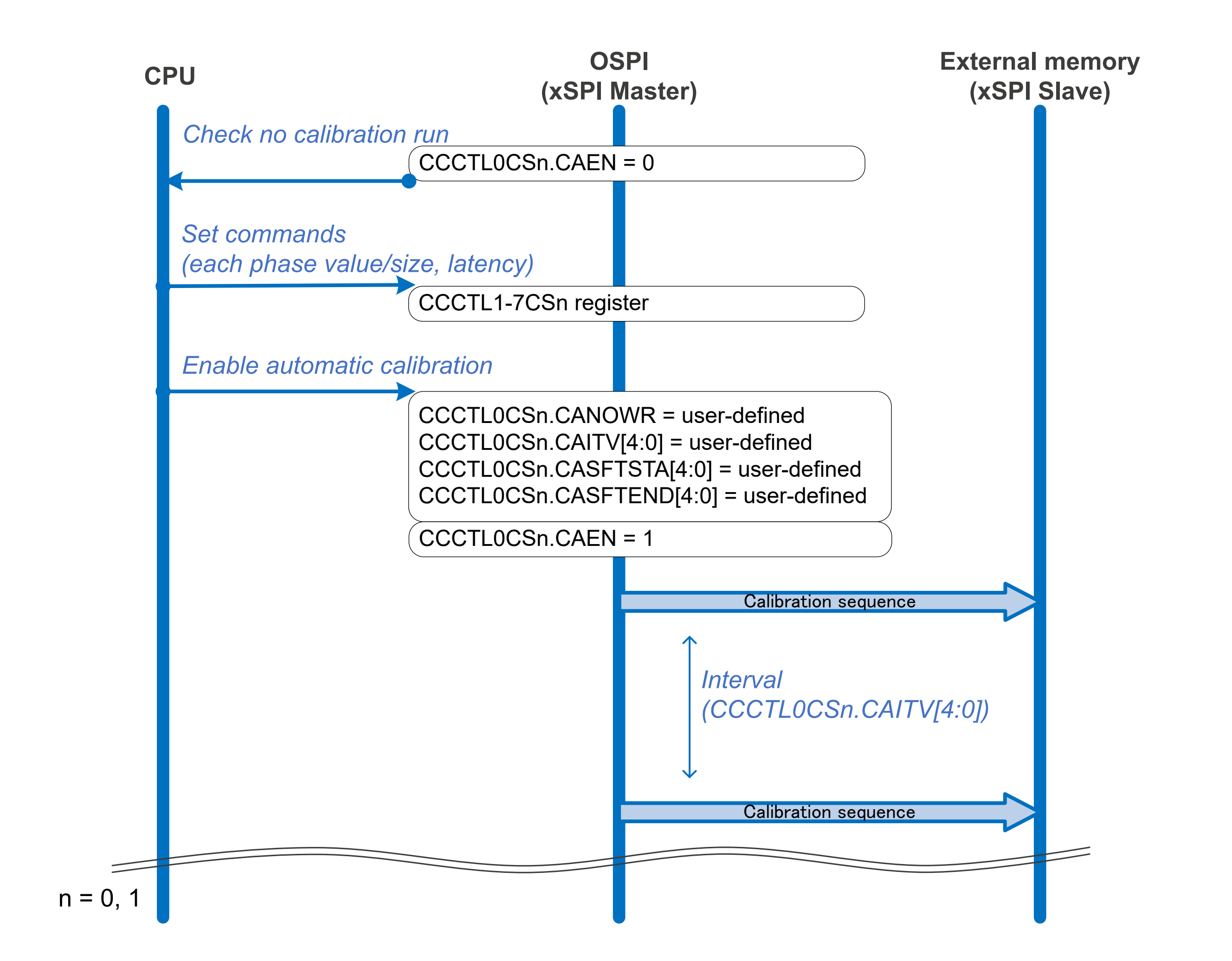


**Figure 45.24 Flow of automatic calibration**

### 45.3.7.4 Flow of Manual-command Procedure

Figure 45.25 shows the manual-command procedure for direct mode.

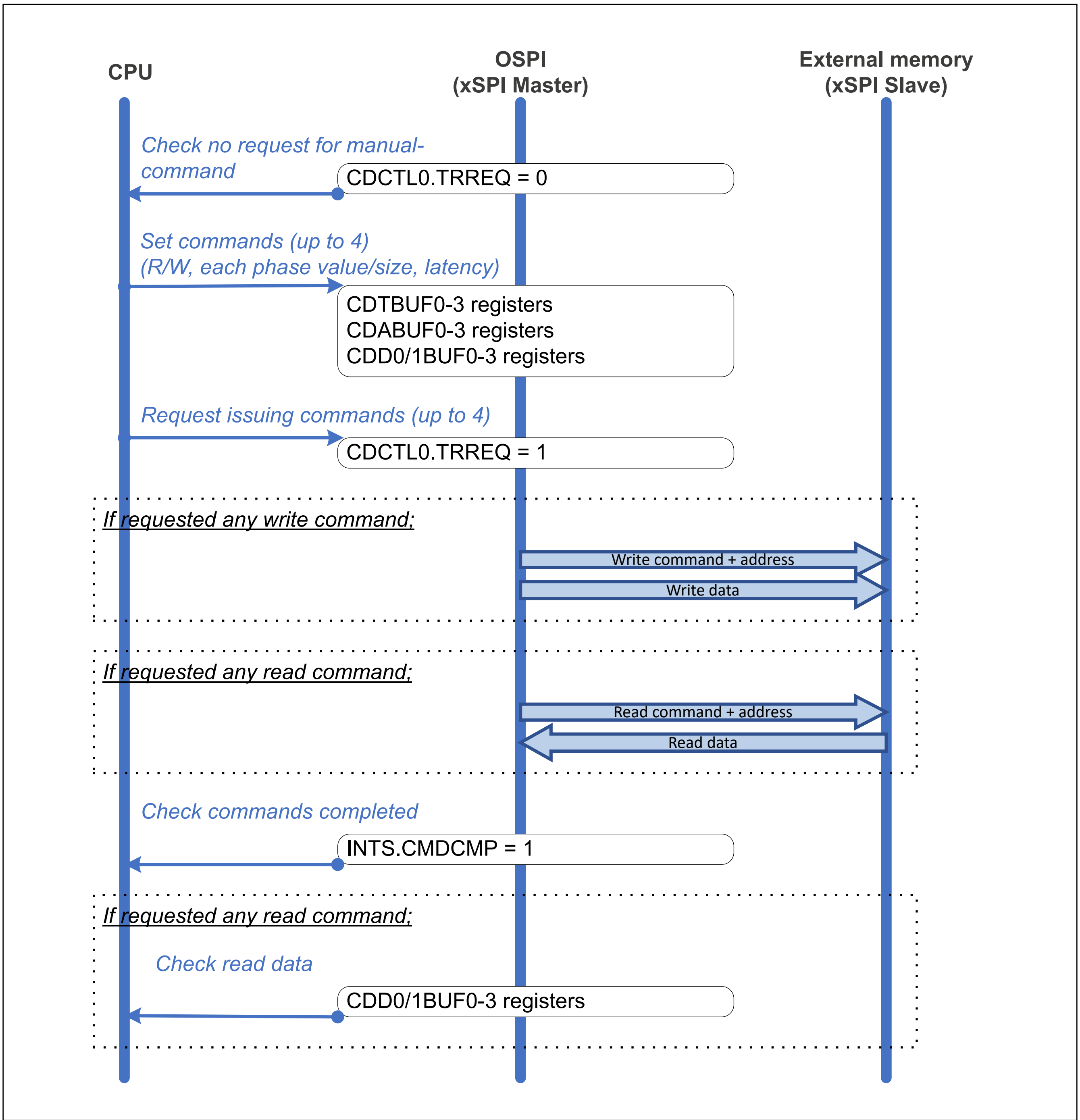


**Figure 45.25 Flow of manual-command procedure for direct mode**

Figure 45.26 shows the manual-command procedure for periodic mode.

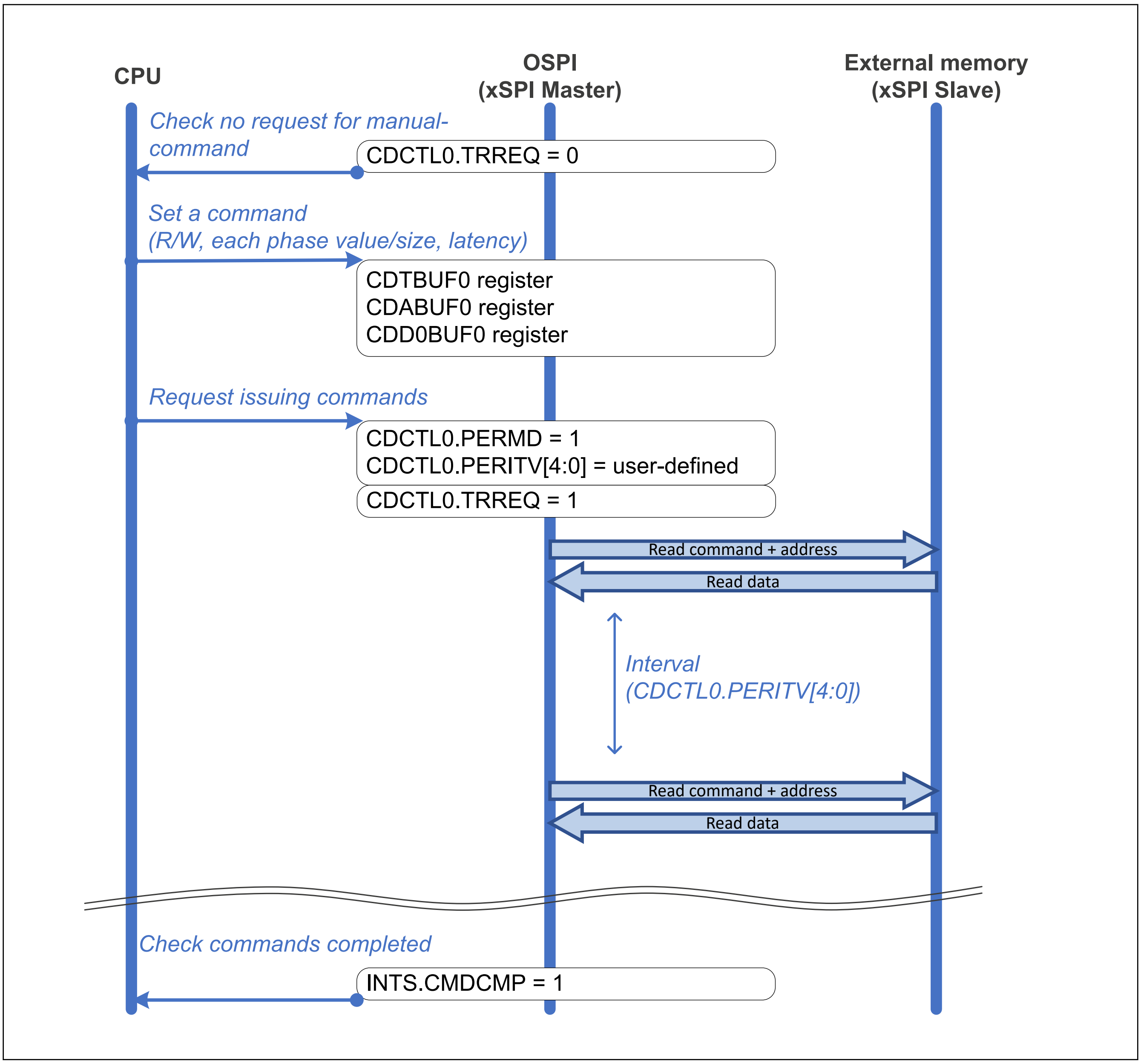


**Figure 45.26 Flow of manual-command procedure for periodic mode**

### 45.3.7.5 Flow of Memory-mapping

Figure 45.27 shows the flow of a memory-mapping.

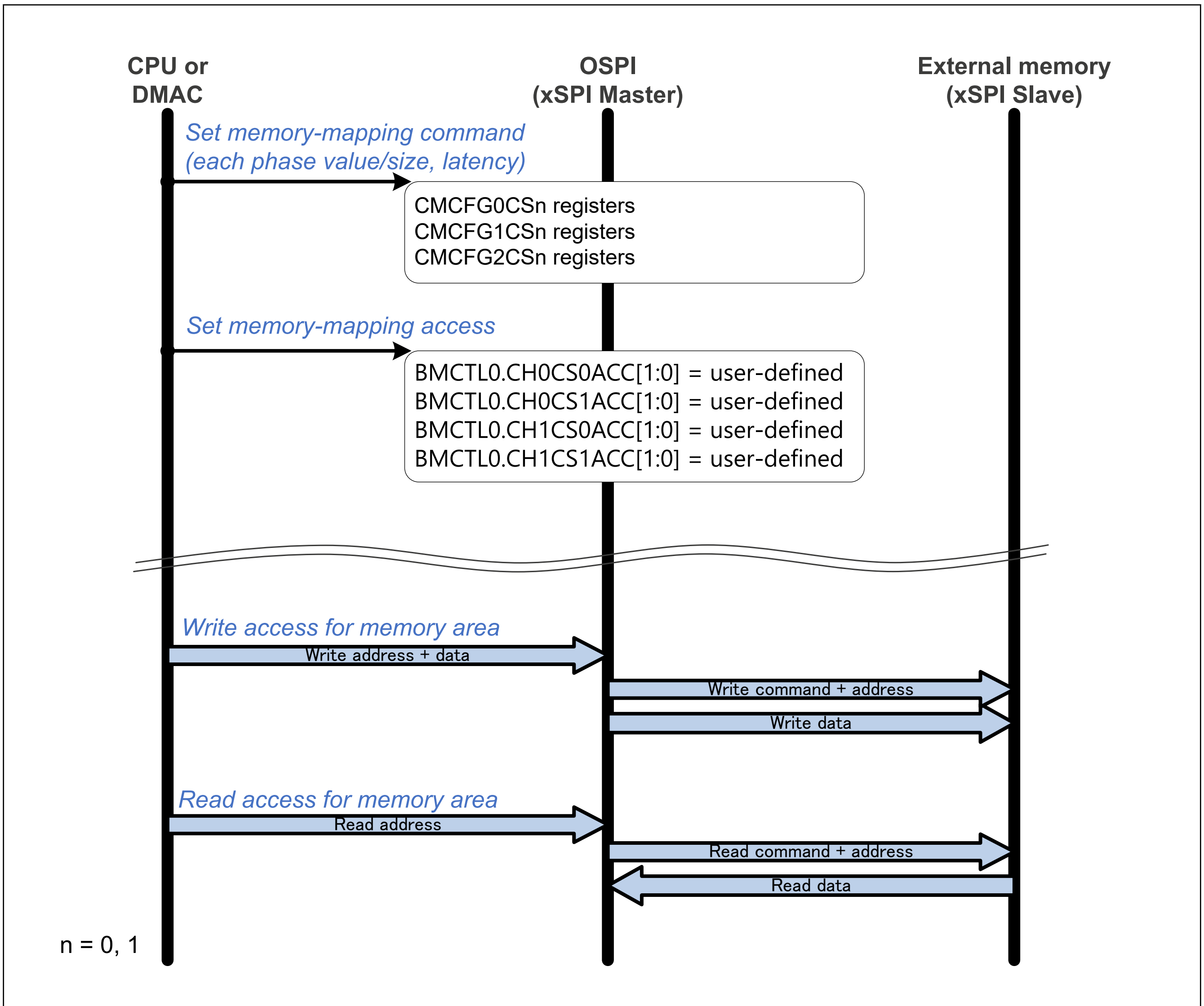


**Figure 45.27 Flow of memory-mapping**

### 45.3.7.6 Flow of Memory-mapping Stop

Figure 45.28 shows the flow of a memory-mapping stop.

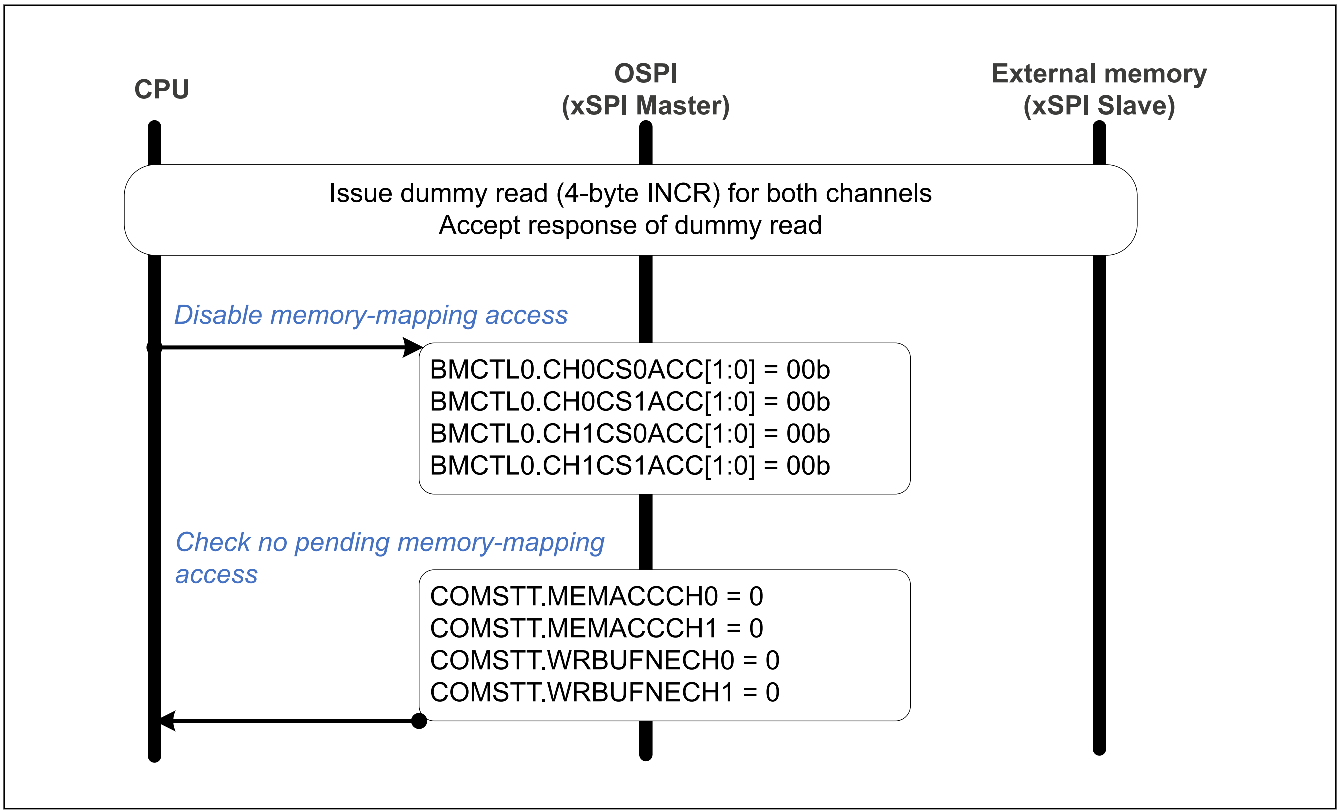


**Figure 45.28 Flow of memory-mapping stop**

### 45.3.7.7 Flow of Pattern Request

Figure 45.29 shows the flow of a pattern request. Before requesting any pattern, any ongoing commands should be completed or canceled.

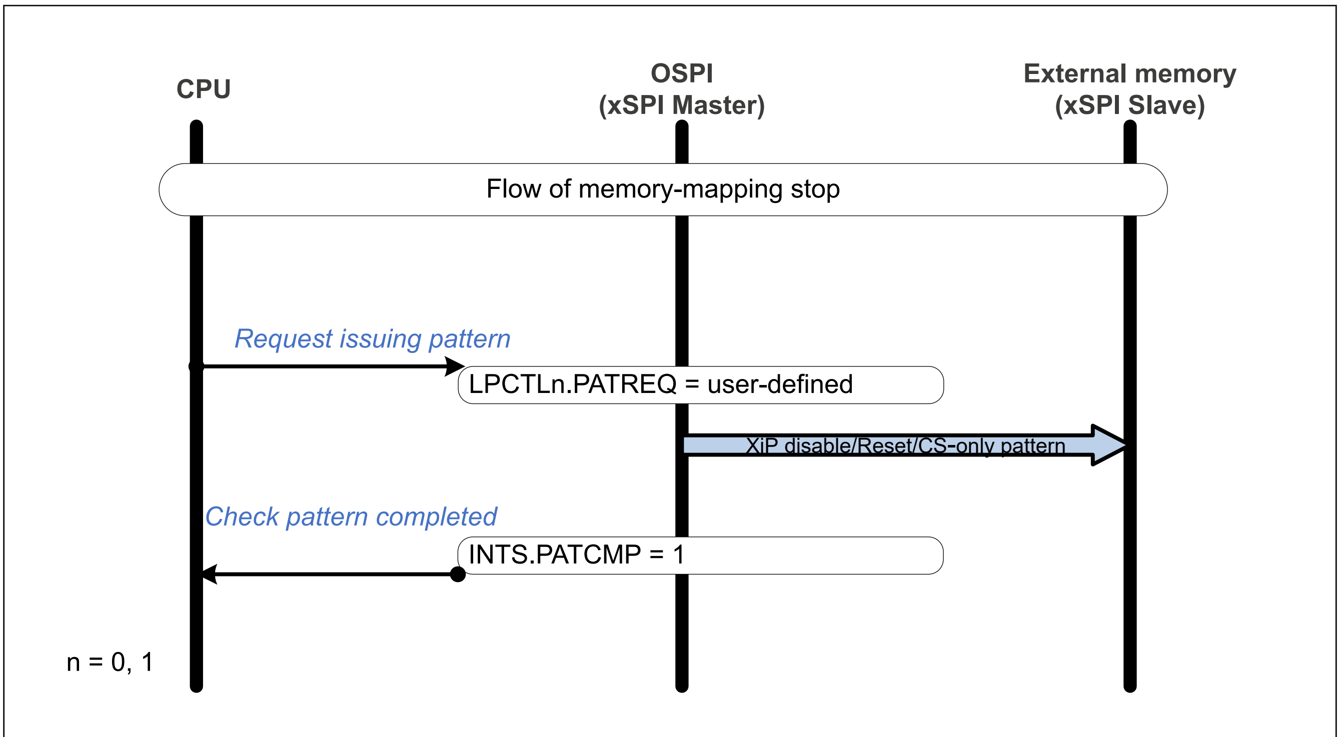


**Figure 45.29 Flow of pattern request**

### 45.3.7.8 Flow of XiP Mode

Figure 45.30 shows the flow of XiP mode enable/disable.

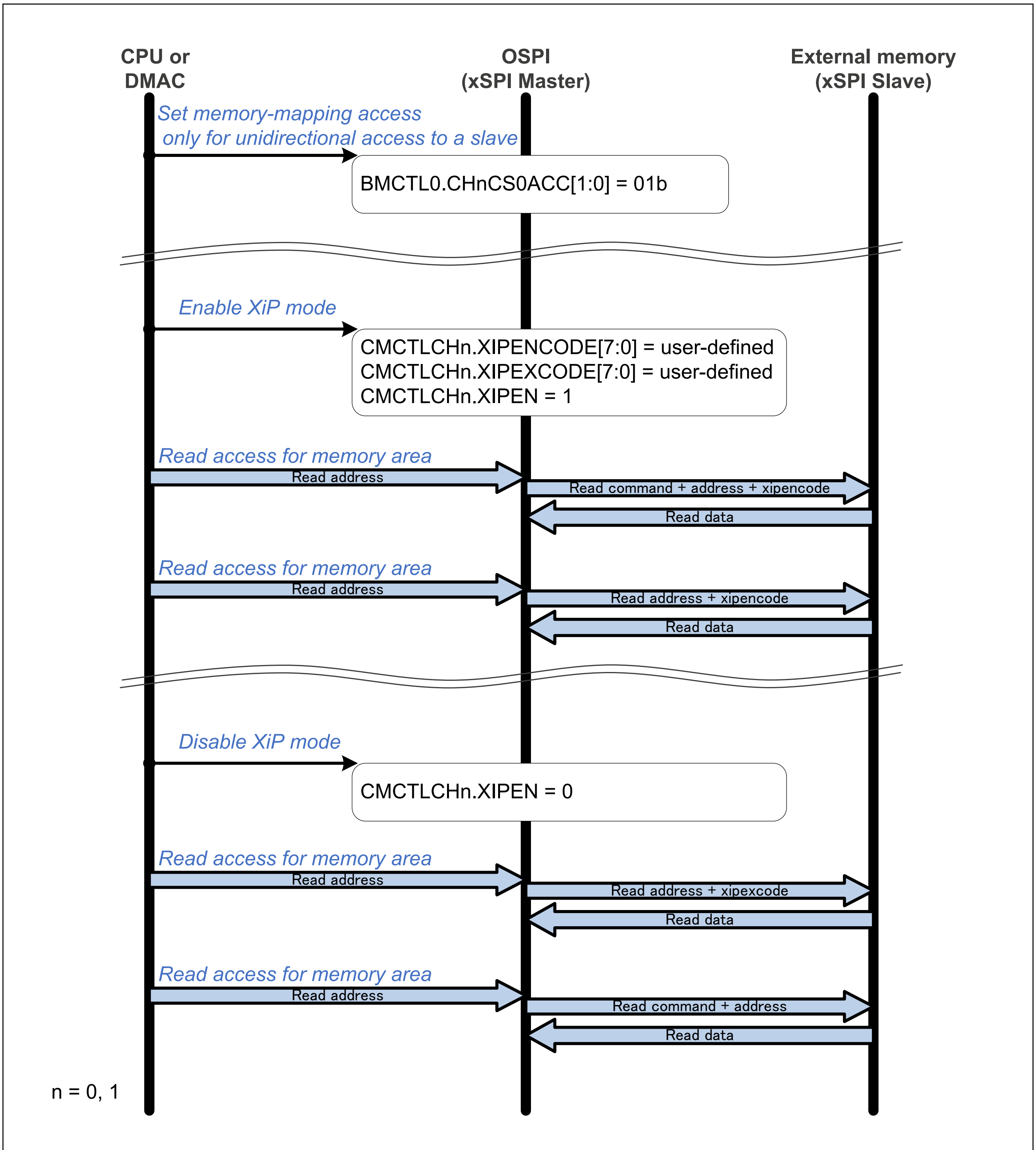


**Figure 45.30 Flow of XiP mode enable/disable**

### 45.3.7.9 Flow of Read While Write

Figure 45.31 shows the example flow of a read while write.

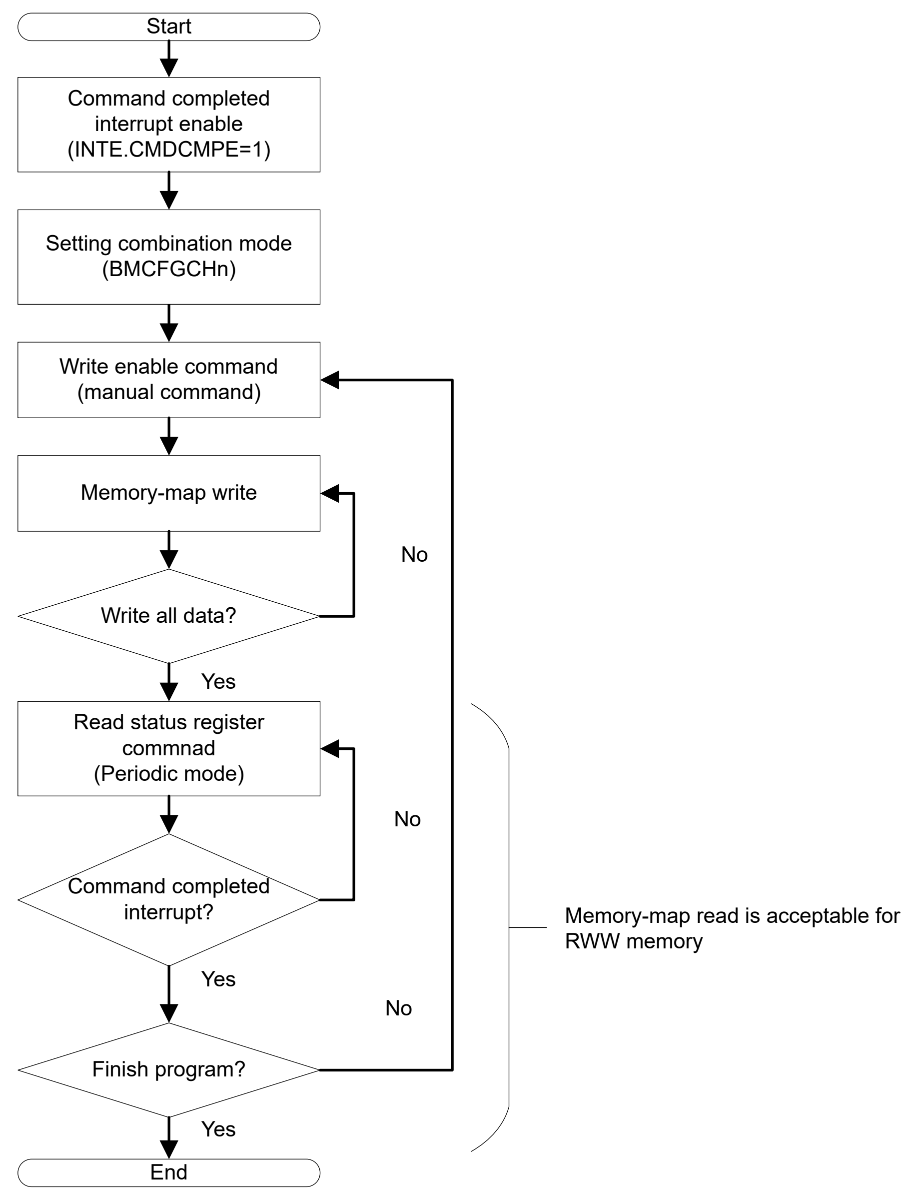


**Figure 45.31 Flow of read while write**

## 45.4 SiP Product Configuration

Flash memory ISSI IS25WX064-JWLE is packaged for SiP product.

- VCC2 = 1.70 V to 2.00 V
- OM_1_SCLK frequency:
  - SDR without OM_DQS: Up to 50 MHz
  - SDR with OM_DQS/DDR: Up to 133 MHz.

For details, see section 70, Electrical Characteristics and the datasheet of IS25WX064.

The following settings should be set before accessing the SiP Flash memory.

- Set OSPI (11100b) to the PmnPFS.PSEL[4:0] bits of P603 to P607 and PC00 to PC07
- Set the LVOCR.LVO1E bit to 1
- Set the PSARB.PSARB17 bit to 0 unless whole SiP Flash area is marked as Non-secure

- Set the end address of CS1
  - Set the RACTL.RAEN bit to 1
  - Set the MER register to 0x0871_0FF0 for 8 MB SiP-Flash product
  - Set the MER register to 0x0831_0FF0 for 4 MB SiP-Flash product
  - Set the RACTL.RAEN bit to 0
- Set the LIOCFGCS1.DDRSMPEX[3:0] bits
  The $t_{CKHDSH}$ value of SiP Flash memory is $t_{PERIOD}/2 + 6$[ns]. When OM1_SCLK is 133 MHz, the DDRSMPEX[3:0] bits should be 0x2.

Figure 45.32 shows connection between the MCU and the SiP Flash memory. The external pull-up should be attached to the PUP pin.

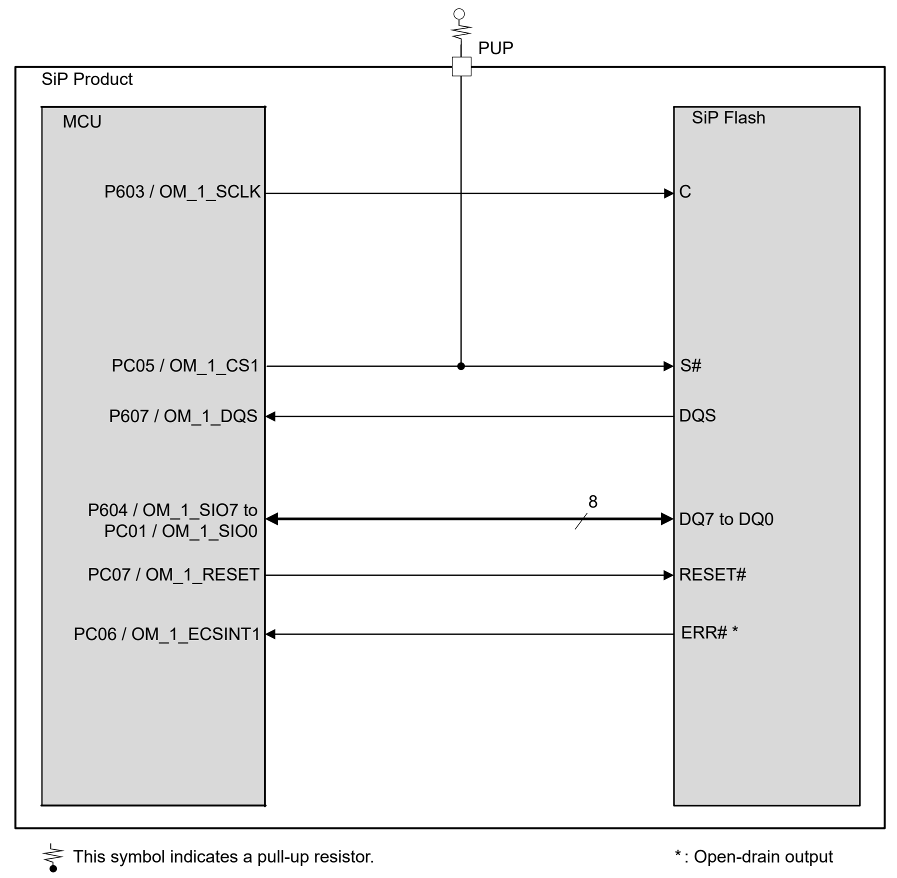


**Figure 45.32 Connection between the MCU and the SiP Flash memory**

## 45.5 Usage Notes

### 45.5.1 Software Standby Mode

Before transition to Software Standby mode, stop communication by Flow of Communication Stop, see section 45.3.7.2. Flow of Communication Stop.

### 45.5.2 Constraints on Memory Access

Memory access has the following constraints:

- Not using the prefetch function if the application has access to the last 128-byte area

- Enable the prefetch function when read HyperFlash by DRW or DMAC/DTC in Block transfer mode or Repeat block transfer mode and source address is Incremented mode

### 45.5.3 Constraint on Burst Length

AXI bus master must have a burst length of 16 or less.

### 45.5.4 Memory Write Combination Mode

When BMCFGCHn.MWRCOMB = 1, the memory write combination mode is enabled. However, the combination does not work if the bus master is CPU0 and OSPI data is under 32 bits. Table 45.11 shows the possibility of a data write for each bus master.

**Table 45.11 Data write possibility for each bus master**

| Bus master | Combination enable | Combination disable |
|---|---|---|
| CPU0 under 32-bit access | Not possible | Possible |
| CPU0 64-bit access | Possible | Possible |
| Other bus masters | Possible | Possible |

### 45.5.5 Module-stop Function

OSPI operation can be disabled or enabled using Module Stop Control Register B (MSTPCRB). The OSPI module is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see section 11, Low Power Mode.

### 45.5.6 Byte Data Order in 8-pin DDR Mode

In 8-pin-DDR mode, OSPI read and write data in the order of byte 0 at the rising edge and byte 1 at the falling edge.

Some memory devices read and write in the swapped order in 8-pin DDR mode. For these devices, if 8-pin DDR read the data which is written in other than 8-pin DDR mode, this data should be written in swapped order.

### 45.5.7 Constraint on Prefetch Function

When all the following conditions are satisfied, prefetch function has no effect on the read access even if the PREEN bit is set to 1.

- Master is CPU0 or CPU1
- Memory attribution of OSPI region is set to cacheable memory by MPU in the CPU
- CPU cache is enabled
- DOTF is enabled.

### 45.5.8 Constraint on Block Protect setting for Flash 4MB SiP Product

Flash memory ISSI IS25WX064-JWLE has block protect function. There are some prohibited settings for Flash 4MB SiP Product (see section section 1.3. Part Numbering).

Table 45.12 shows block protect setting of TB and BP[3:0] bits in Status Register.

**Table 45.12 Block protect setting for Flash 4MB SiP Product (1 of 2)**

| Status Register Bits | | | | | Byte Protected |
|---|---|---|---|---|---|
| TB | BP3 | BP2 | BP1 | BP0 | |
| 1 | 0 | 0 | 0 | 0 | 0 KB |
| 1 | 0 | 0 | 0 | 1 | 128 KB |
| 1 | 0 | 0 | 1 | 0 | 256 KB |
| 1 | 0 | 0 | 1 | 1 | 512 KB |
| 1 | 0 | 1 | 0 | 0 | 1 MB |

**Table 45.12 Block protect setting for Flash 4MB SiP Product (2 of 2)**

| Status Register Bits | | | | | Byte Protected |
|---|---|---|---|---|---|
| TB | BP3 | BP2 | BP1 | BP0 | |
| 1 | 0 | 1 | 0 | 1 | 2 MB |
| 1 | 0 | 1 | 1 | 0 | 4 MB |
| Others: Setting prohibited | | | | | |

## 45.5.9 Constraint on OTP Area for SiP Product

Flash memory ISSI IS25WX064-JWLE has 64-byte OTP area, but it is not supported in the SiP product.